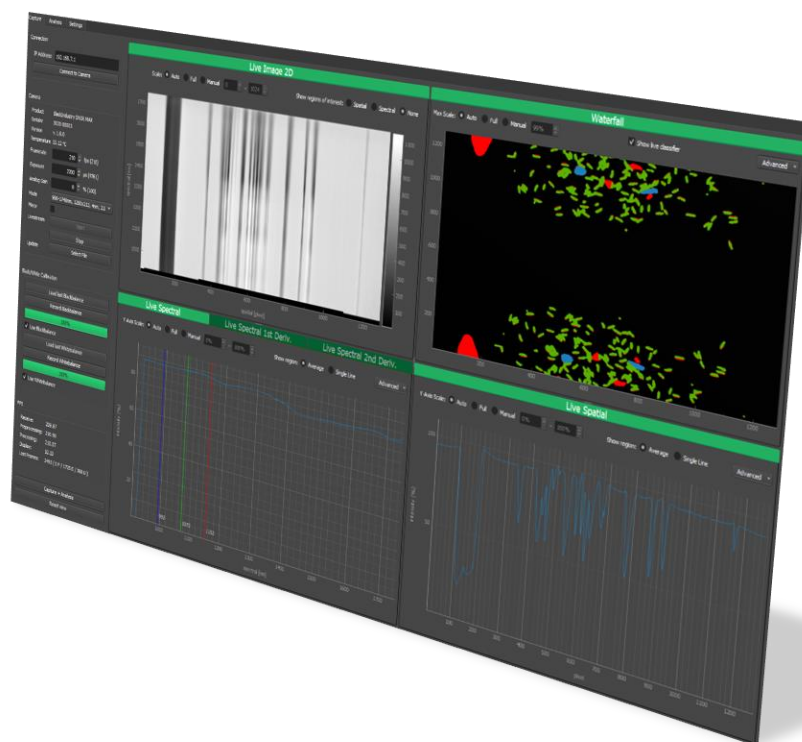




User Manual

HAIP BlackStudio Software



1 INTRODUCTION

Dear Customer,

We would like to thank you for the purchase of a HAIP Solutions GmbH product and wish you successful, insightful and at the same time pleasant work.

HAIP BlackStudio is a recording software for hyperspectral images that enables the acquisition of real-time spectral data. The software is dedicated to the series of BlackIndustry cameras for industrial in-line applications.

1.1 ABOUT THE USER MANUAL

Please read this User Manual carefully before first use and study the warnings and safety instructions. Keep the instructions! This allows you to find and look up important clues later.

We hope that the User Manual answers all your questions. If this is not the case or if you would like to report a service or warranty claim, please contact our support team via our website:

www.haip-solutions.com

Or contact us directly:

Corporate headquarters

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Phone: +49 511 37352860
E-mail: info@haip-solutions.com
Web: www.haip-solutions.com

1.2 ABOUT HAIP SOLUTIONS

HAIP Solutions GmbH is a driver of innovation in the field of hyperspectral camera solutions. It was founded in October 2019 and is based in Hanover.

We develop, produce, and distribute high-quality and user-oriented hyperspectral systems for applications in the fields of agriculture & forestry, food industry and recycling.

With our application-oriented product solutions, we provide competent and flexible support for both commercial and research applications.

In addition to stationary hyperspectral imaging systems, our products also include solutions for mobile applications, such as handheld or UAV-based.

1.3 DISCLAIMER AND PERFORMANCE CHARACTERISTICS OF THE SYSTEM

HAIP Solutions GmbH can neither guarantee nor check the performance characteristics of the software and rejects any warranty and liability claims if products of HAIP Solutions GmbH have been misused, neglected, or damaged. Furthermore, this applies if the products have been damaged by external influences or have not been used in accordance with the enclosed instructions for use; if accessories that are not recommended by HAIP Solutions GmbH have been used, or if repair work has been carried out by service personnel not authorized by HAIP Solutions. The performance of the system may vary depending on several factors, including but not limited to hardware and network infrastructure used, the complexity and size of data being processed and external factors such as internet connectivity and server availability. We do not guarantee that the system will be free from bugs, errors, or vulnerabilities. We will make continuous efforts to address any reported issues and provide updates as necessary.

1.4 PATENTS/REGISTERED TRADEMARKS/COPYRIGHT

This operating manual is the property of HAIP Solutions GmbH. A complete or partial duplication of the instructions as well as the transfer to third parties are not permitted without the express written consent of HAIP Solutions GmbH.

HAIP Solutions GmbH makes every effort to ensure the accuracy of the information in the operating instructions. This document is subject to change without notice.

CONTENT

1	INTRODUCTION	2
1.1	ABOUT THE USER MANUAL.....	2
1.2	ABOUT HAIP SOLUTIONS.....	3
1.3	DISCLAIMER AND PERFORMANCE CHARACTERISTICS OF THE SYSTEM	3
1.4	PATENTS/REGISTERED TRADEMARKS/COPYRIGHT	3
2	RECENT UPDATES / CHANGE LOG	7
2.1	VERSION 0.2.144	7
2.2	VERSION 0.2.143	7
2.3	VERSION 0.2.139	8
2.4	VERSION 0.2.85	8
2.5	VERSION 0.2.77	8
2.6	VERSION 0.2.73	8
2.7	VERSION 0.2.46	9
2.8	VERSION 0.2.44	9
2.9	VERSION 0.2.42	9
2.10	VERSION 0.2.36	9
3	ADDITIONAL TUTORIALS	10
4	HAIP BLACKSTUDIO SOFTWARE – OVERVIEW.....	10
5	BLACKSTUDIO – CAPTURE TAB	10
5.1	CAPTURE TAB – OVERVIEW.....	10
5.2	CAPTURE TAB - LIVE IMAGE 2D WINDOW	11
5.3	CAPTURE TAB - WATERFALL WINDOW	13
5.4	CAPTURE TAB - LIVE SPECTRAL WINDOW	14
5.5	CAPTURE TAB - LIVE SPATIAL WINDOW.....	15
5.6	CAPTURE TAB - CAMERA SETTINGS.....	16
5.7	CAPTURE TAB – BLACKBALANCE / WHITEBALANCE	18
5.8	CAPTURE TAB – FPS MONITORING.....	20
5.9	CAPTURE TAB – DATA RECORDING	20
6	BLACKSTUDIO - ANALYSIS TAB	21

6.1	ANALYSIS TAB – OVERVIEW	21
6.2	ANALYSIS TAB – CUBE 2D VIEW	22
6.3	ANALYSIS TAB – CANVAS VIEW	25
6.4	ANALYSIS TAB - SPECTRAL VIEW	28
6.5	ANALYSIS TAB – SIDEBAR	32
6.5.1	SIDEBAR – DATA AND ROI	32
6.5.2	SIDEBAR – CUBE 2D ANALYSIS	34
6.5.3	SIDEBAR – MEAN/BACKGROUND/CLASSIFIER	35
7	BLACKSTUDIO – DATABASE	40
7.1	DATABASE – OVERVIEW	40
7.2	DATABASE – SNIPPETS COLLECTION	41
7.3	DATABASE – CATEGORIES	42
7.4	DATABASE – FILTER	44
8	BLACKSTUDIO – STATISTICS TAB	45
8.1	STATISTICS TAB – DETECTED MATERIALS HISTOGRAM	46
8.2	STATISTICS TAB – DETECTED MATERIALS TABLE	46
9	BLACKSTUDIO – SETTINGS TAB	47
10	DATA ACQUISITION IN BLACKSTUDIO	50
10.1	DATA ACQUISITION WORKFLOW	50
10.2	DATA ACQUISITION – PIXEL PACKING	54
10.3	DATA ACQUISITION - EXTERNAL TRIGGER FUNCTIONALITY	54
10.3.1	EXTERNAL TRIGGER - FRAME START MODE	56
10.3.2	EXTERNAL TRIGGER - FRAME BURST MODE	56
10.3.3	EXTERNAL TRIGGER - TRIGGER OUTPUT	57
10.4	DATA ACQUISITION – MULTI ROI MODES	58
10.4.1	FUNCTIONALITY OF SPECTRAL REGIONS	58
10.4.2	SELECT SPECTRAL REGIONS IN BLACKSTUDIO	60
11	DATA ANALYSIS IN BLACKSTUDIO	62
11.1	DATA ANALYSIS – SETTING SPATIAL ROI MARKERS	62
11.2	DATA ANALYSIS – PREPARE DATABASE AND CREATE A SNIPPET	63
11.3	DATA ANALYSIS – LOAD SNIPPETS TO CANVAS	65
11.4	DATA ANALYSIS – TRAIN A CLASSIFIER	66
11.4.1	GENERAL CLASSIFICATION WORKFLOW	66
11.4.2	BACKGROUND FILTER	69
11.4.3	CLASSIFICATION ALGORITHMS	71

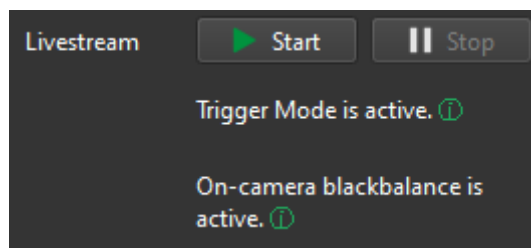
11.4.4	REGRESSION	72
11.4.5	VISUALIZATION	73
11.4.6	VISUALIZE WAVELENGTH IMPORTANCE.....	75
11.4.7	MULTIPLE CLASSIFIERS	75
11.5	DATA ANALYSIS – ON CAMERA CLASSIFIER.....	76
11.5.1	ON-CAMERA CLASSIFIER OUTPUT MODE	76
11.5.2	ON-CAMERA CLASSIFIER IN FULL SPECTRAL RANGE:	77
11.5.3	ON-CAMERA CLASSIFIER IN MULTI-ROI MODE USING SPECTRAL REGIONS:.....	79
12	TROUBLESHOOTING	83
13	ACCESS TO HAIP BLACKSTUDIO SOFTWARE	83
14	SAMPLE DATA	83
15	MAINTENANCE AND SERVICE	84
16	SYSTEM REQUIREMENTS.....	84

2 RECENT UPDATES / CHANGE LOG

2.1 VERSION 0.2.144

New features:

- On-camera classifier is loaded and transferred with Project load/save (see chapter 9)
- GUI for recording a specific number of lines (see chapter 5.9)
- PCA Visualization (see chapter 11.4.5)
- Load Wavelength Table (see chapter 6.4)
- New pre-processing dialog (see chapter 6.5.3)
- Mixed class colors (see chapter 6.5.3)
- Statistics tab available (see chapter 8)
- Newly added warnings in the Capture tab when trigger functionality and/or on-camera blackbalance are enabled:



New Bug Fixes:

- Fixing disconnect issues
- Fixing exposure bug
- Fixing segmented White balance
- Fixing On-camera blackbalance issue: **Be aware that our last BlackIndustry SWIR update could have enabled On-Camera blackbalance by accident.**

2.2 VERSION 0.2.143

New features: Some of the new features are only available with the new BlackIndustry SWIR 1.7 camera version v.1.1.18:

- Class Names are saved with the On-Camera-Classifer and can be received from the camera
- Multiple Classifiers can be collected and transferred to the camera to retrieve multiple classifications at once
- The current state of BlackStudio and the camera can be saved in a complete project that can be loaded to restore the complete state of BlackStudio and the camera
- Exporting of canvas as *.png
- Option to rearrange the canvas to fit the horizontal width of the current zoom
- Option to scroll canvas vertically using CTRL+WHEEL

- Option to zoom on single snippets in Canvas
- Option to save Snippet Positions in a database Snippet Group
- Disabling Unknown Class detection is now Standard
- Renaming of blackbalance and whitebalance files
- Temperature values are saved and displayed with HSI captures
- The number of spectral bands is now displayed when selecting Spectral Regions
- Preparing for official Linux Support
- User Manual Update

New Bug Fixes:

- Connection failing even when the camera was connected
- Applying the Background Filter after the Regression resulted in the wrong Foreground Color
- Disabled OpenGL because it was introducing a lot of graphical bugs and crashes
- Error Message when .img file size does not match its header information

2.3 VERSION 0.2.139

- Performance Improvements
- Support für Lazy Loading of HSI data (turn on in Settings)
- Bug Fixes
- Little GUI Improvements
- More classifiers
- Feature Importance Visualization

2.4 VERSION 0.2.85

- Allows different spectral resolutions and ranges to be loaded together
- Many smaller bug fixes

2.5 VERSION 0.2.77

- Fixed crop when rois are not inside crop
- Fixed crop when region selection is applied multiple times

2.6 VERSION 0.2.73

- Many bug fixes
- Applying mean filter beforehand to visualize and save time on classification
- Signed datatype support - blackbalance can reduce values < 0
- Auto updater

2.7 VERSION 0.2.46

- Allow to configure background to be all pixel values above a threshold instead of below
- Allow to write on-camera-classifier transfer files to file to transfer them later in BlackStudio or API

2.8 VERSION 0.2.44

- Fixing triggerDialog crash stream bug
- Adding OneClassSVM Support for One Class Classifications
- Enabling SVM Dialog to select SVM instead of LDA
- Adding progress bars to classification

2.9 VERSION 0.2.42

- New .png export option for cube2d
- Option for segmented white balance
- Fixing a bug resulting in fps could not load correctly even when „Save config“ was set
- Fixing a bug that image2d view colorbar was not filled with the colormap
- Fixing on-camera classifier transfer on classifier trained on data captured with mode 3

2.10 VERSION 0.2.36

- Fix ALT-TAB resulting in spectral mouse preview being stuck
- Fix bug with loading original row names from db to canvas
- Image cube 2d manual scale
- Fix saving png to network location
- Fix bug resulting in unloadable .hdr files
- New database features
 - Add new entries for fields when adding a new snippet
 - Create DB materials with color that can be chosen in analysis view using \$
 - Verified_by is now dropdown

3 ADDITIONAL TUTORIALS

Coming soon ...

4 HAIP BLACKSTUDIO SOFTWARE – OVERVIEW

The **HAIP BlackStudio** software is a data acquisition, processing and analysis software designed for the hyperspectral cameras of the **HAIP BlackIndustry** series.

The software is divided into five different main tabs:

- **Capture Tab** to perform and overview the data acquisition (chapter 5)
- **Analysis Tab** to process and analyze the acquired data (chapter 6)
- **Database Tab** for permanently managing and storing data (chapter 7)
- **Statistics Tab** for observing materials statistics. Deactivated by default (chapter 8)
- **Settings Tab** to select basic settings (chapter 9)
- **Manuals Tab** for looking up the User Manual and Quick Start Guide.

The following chapters introduce the individual tabs and explain their scope and functions.

5 BLACKSTUDIO – CAPTURE TAB

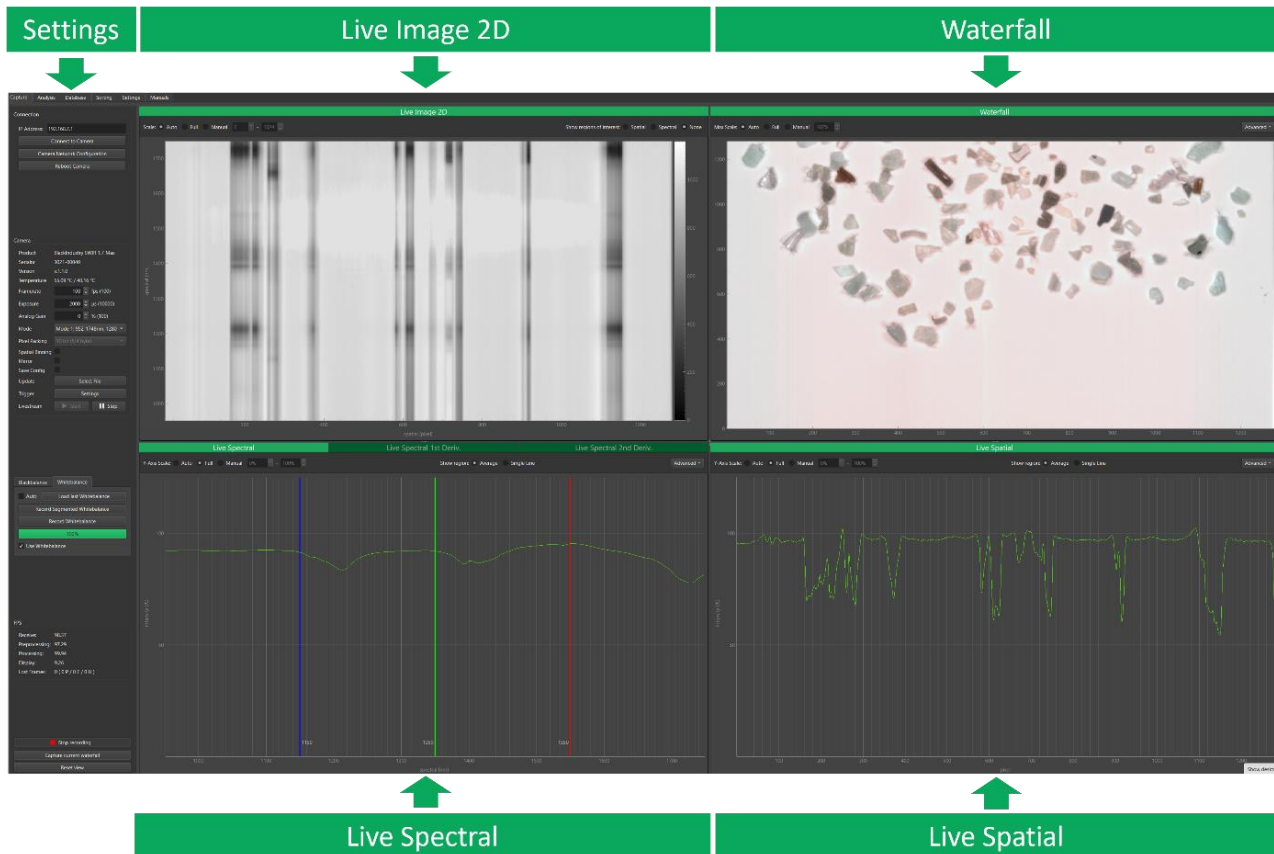
In the **Capture tab** of the HAIP BlackStudio software, all settings for successful data acquisition can be adjusted. Additionally, information about the currently captured hyperspectral data is displayed via the four **livestream windows**:

- **Live Image 2D**
- **Waterfall**
- **Live Spectral**
- **Live Spatial**

5.1 CAPTURE TAB – OVERVIEW

The **Capture tab** consists of four livestream windows and a settings section on the left-hand side.

- The livestream windows, indicated by the green title bar, can be freely arranged by drag & drop or resized using the mouse cursor.
- Open the individual windows to full screen by double clicking on the dedicated green title bar.
- Via the **Reset View** button, the default arrangement of the windows can be regained.



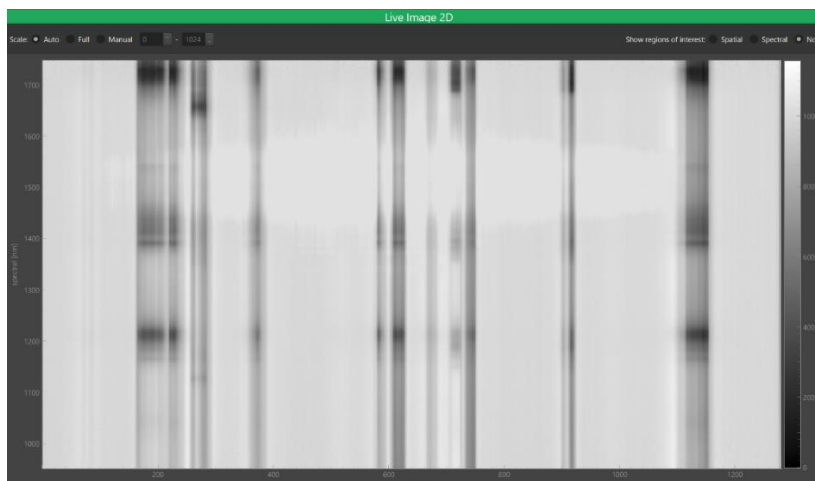
5.2 CAPTURE TAB - LIVE IMAGE 2D WINDOW

Displays currently recorded pixel line (Spatial x Spectral).

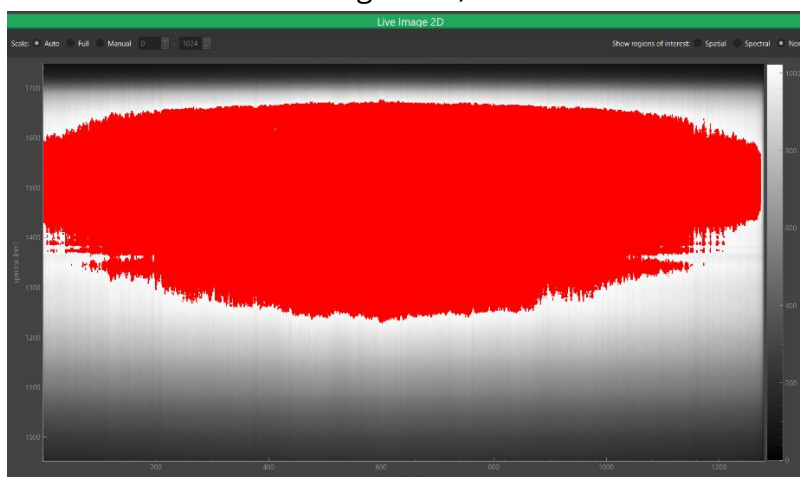
Settings:

- **Scale:** Adjust the intensity scale [Auto adjustment | Full stretch | Manual selection].
- **Show regions of interest:** Show and adjust regions of interests (ROIs) [Spectral | Spatial | None]: The intensity information of the current recording is obtained from these ROIs (see **Live Spectral** & **Live Spatial**).
 - Left Double click to open a new ROI
 - Right-click to delete a ROI
 - Adjust the ROI width by sliding the region markers
 - Click on a ROI and hold to move an entire ROI
 - Note that several ROIs can be defined at the same time (see screenshot below).
- **Advanced: Export .png:** Export currently displayed Live Image 2D View as *.png file.

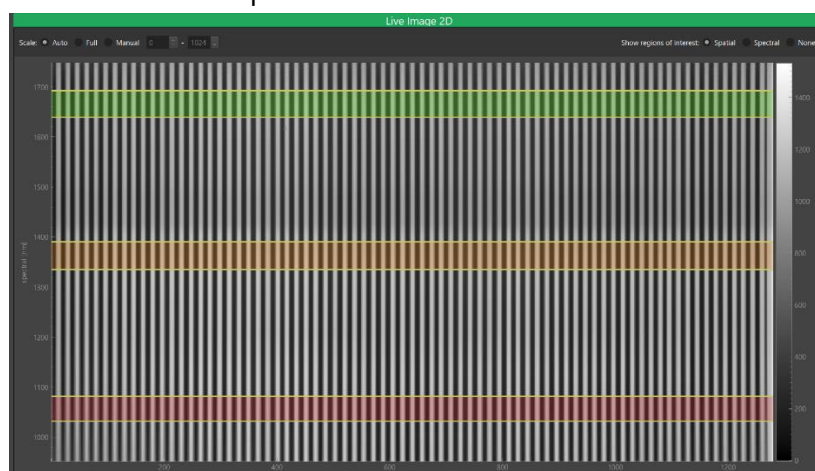
Live Image 2D View with active Livestream:



Live Image 2D View with active Livestream, showing oversaturation in red: Adjust Exposure and Analog Gain settings to achieve an adequate intensity level and to eliminate oversaturation (only available in 10-bit Pixel Packing Mode):



Live Image 2D View with several spatial ROIs:

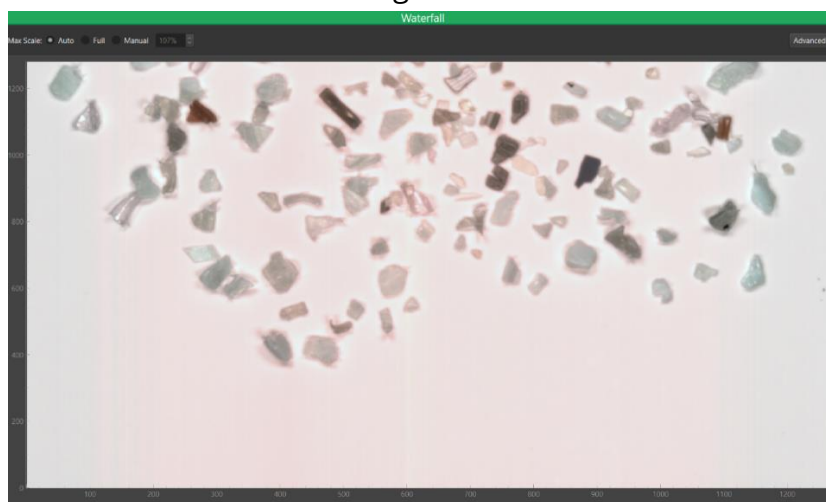


5.3 CAPTURE TAB - WATERFALL WINDOW

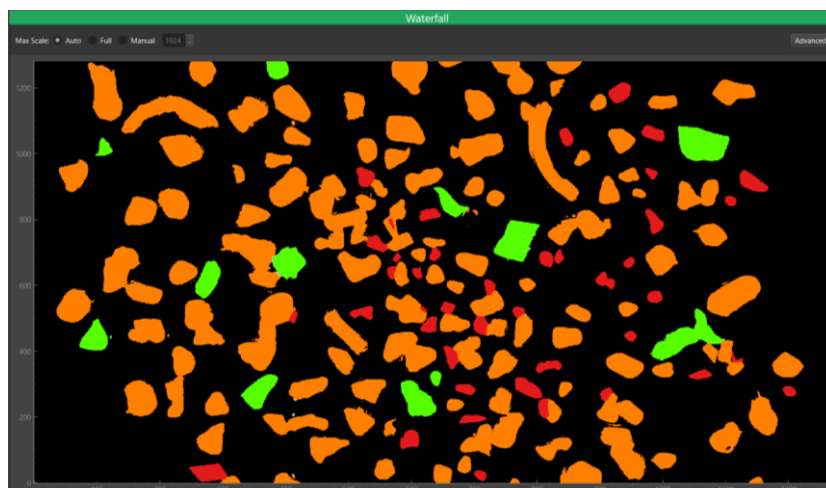
All recorded pixel lines are displayed consecutively. The latest line is added at the top of the screen (Spatial x Spatial).

Settings:

- **Max Scale:** Adjust the intensity scale of the displayed data [Auto adjustment | Full stretch | Manual selection].
- **Advanced: Enable live classifier** for live classification as soon as a classifier has been trained.
 - The live classifier enables the live classification of the recorded objects after a preliminary image classification.
 - This function is still available but is **deprecated**. It is recommended to use the new **On-Camera Classifier** tool (see chapter 11.5).
- **Advanced: Export.png:** Export currently displayed Waterfall View as *.png file.
- By moving the red, green and blue sliders in the **Live Spectral View**, the color scheme in the **Waterfall window** can be changed.



Classified Waterfall window via on-camera classifier:

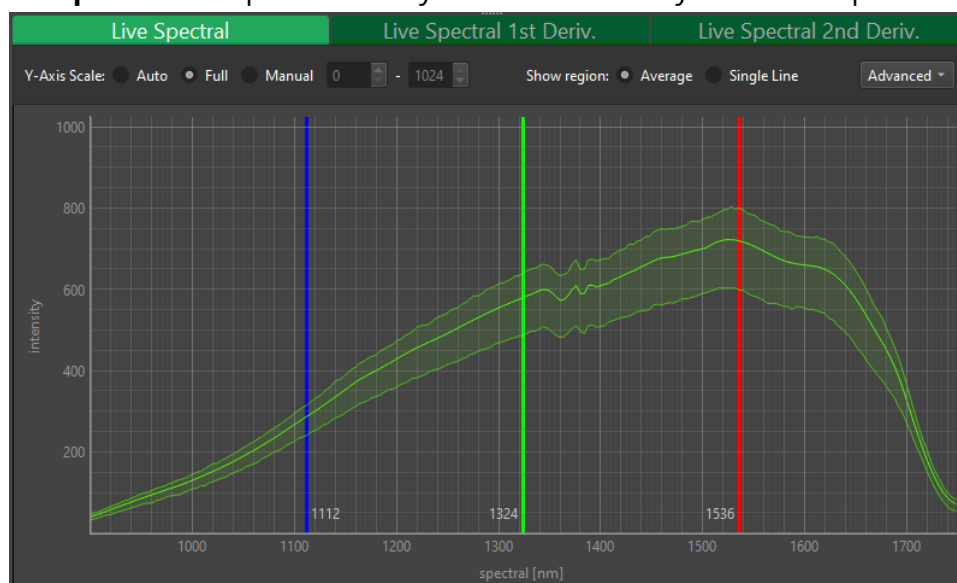


5.4 CAPTURE TAB - LIVE SPECTRAL WINDOW

Displays the spectral intensity for each camera wavelength (Spectral x Intensity).

Settings:

- By moving the red, green and blue sliders, the color scheme in the Waterfall window can be changed.
- **Y-Axis scale:** Auto | Full | Manual (identical to Live 2D View).
- **Show region:**
 - **Average:** Average value of the intensity of the spectral ROI. Values from several ROIs are color-coded.
 - **Single Line:** Intensity values of the single middle line of a spectral ROI. Values from several ROIs are color-coded.
- **Advanced:**
 - **Use mono slider:** one greyscale slider for the color scheme of the Waterfall window.
 - **Show Min/Max:** Displays intensity value range from min to max.
 - **Min/Max Transparency:** Activated by default. Displays the min-max range in color. Otherwise, only single lines are displayed for the min and max values. Caution: Enabling this may impact performance.
 - **Export .png:** export current view as *.png file.
 - **Export .csv:** export currently recorded intensity values of a pixel line.
- **Live Spectral 1st Derivation & 2nd Derivation:** Display the first/second derivation of spectral intensity (spectral x intensity).
 - **Savitzky-Golay:** Perform Savitzky-Golay Smoothing.
 - **Export .png:** export current view as *.png file.
 - **Export .csv:** export currently recorded intensity values of a pixel line.



5.5 CAPTURE TAB - LIVE SPATIAL WINDOW

Displays the spectral intensity for each pixel line (Spatial x Intensity).

Settings:

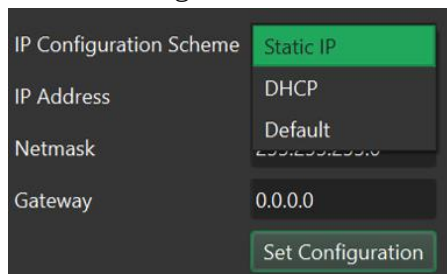
- **Y-Axis scale:** Auto | Full | Manual (identical to Live 2D View).
- **Show region:**
 - **Average:** Average value of the intensity of the spatial ROI. Values from several ROIs are color-coded.
 - **Single Line:** Intensity values of the single middle line of a spatial ROI. Values from several ROIs are color-coded.
- **Advanced:**
 - **Show Min/Max:** Display value range from min to max.
 - **Min/Max Transparency:** Activated by default. Displays the min-max range in color. Otherwise, only single lines are displayed for the min and max values. Caution: Enabling this may impact performance.
 - **Export .png:** export current view as image file.
 - **Export .csv:** export currently recorded intensity values of a pixel line.



5.6 CAPTURE TAB - CAMERA SETTINGS

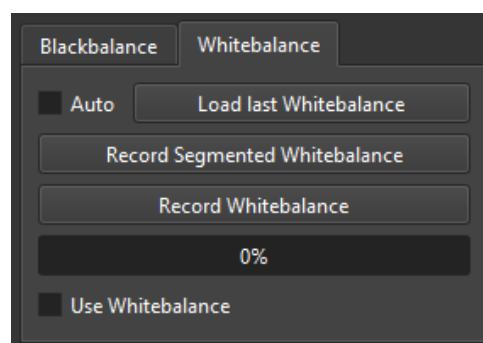
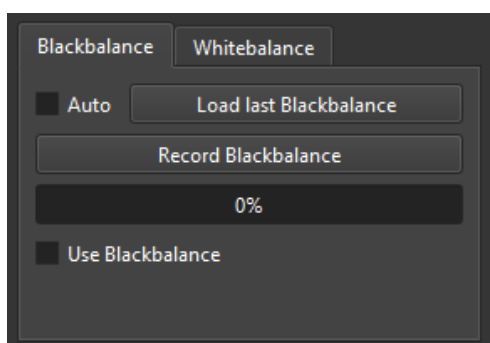
The **Camera settings** section is displayed on the left side of the **Capture tab**. Detailed information about each setting can be found in the table below.

CAMERA SETTINGS	
SETTING NAME	DESCRIPTION
Connection	Enter the IP-address of the camera (default 192.168.7.1) and press Connect to Camera . A status indicator below shows whether the connection was successful or failed.
Camera Info	Once the camera is connected, the following camera information is displayed: • Product (BlackIndustry name) • Serial Number • Version (Firmware version)
Temperature	Displayed when livestream starts. • Sensor/Spectrometer/Processor CAUTION: The sensor temperature must reach 55°C (temperature display changes from red to green when temperature is stabilized) before recording may begin. Temperature 55.06 °C / 34.57 °C / 44.50 °C
Framerate	In frames per second (fps). Select the frame rate for line scan data acquisition. FPS Limit indicates the current maximum fps that can be set, depending on the readout mode.
Exposure	In μ s. The longer the exposure time, the more photons are received by detector, resulting in increased pixel intensity and a “brighter” image (maximum depends on set framerate).
Analog Gain	In %. Increases sensitivity of the sensor, but also noise (maximum 100 %).
Mode	Select a preferred readout mode that is available for the connected camera model. More information in camera user manuals. Exemplary modes for BlackIndustry SWIR 1.7 Max: Mode 0: 952-1748nm, 1280x200, 4nm, 190FPS Mode 1: 952-1748nm, 1280x399, 2nm, 100FPS Mode 2: 952-1748nm, 1280xROI, MULTI-ROI, 1330FPS Mode 3: 952-1748nm, 1280xROI, CALIBRATED-MULTI-ROI, 1330FPS
Pixel Packing	Select a preferred data size (8-bit / 10-bit). The 10-bit data can be packed for the transmission process (5/4 or 3/2) to utilize the

	Ethernet line efficiently. Alternatively select Classified Color Mode , Regression Float or Classified + Spectral . More information see chapter 10.2.
Spatial Binning	Data from the BlackIndustry SWIR 1.7 Max camera is binned to 640 pixels when activated.
Mirror	Depending on the mounting direction of the camera, it is possible to mirror the waterfall and live image by activating the checkbox (Important: Black/White Calibration needs to be renewed after activation).
Save Config	Activate the checkbox to save the settings on camera. The settings are automatically reloaded when the camera is rebooted.
Livestream	Start: Livestream will be displayed on all four windows and the current temperature of the sensor will be automatically received. Stop: stops the livestream of the connected camera.
ADVANCED CONTROL SETTINGS	
Select Update File	Once a new camera firmware version is available, it can be updated via the Select Update File button. - The connected camera must be rebooted for the update to be applied. - All updates must be carried out one after the other and no version number may be skipped. - A previous version can no longer be accessed.
Trigger Settings	If the connected camera supports external trigger functionality, two trigger modes may be selected: - Frame Start - Frame Burst Start For more information see chapter 10.3.
Camera Network Configuration	Change the network configuration according to your needs. Caution: this is a permanent change to the camera. 
Reboot Camera	Manually restart the connected camera. Necessary after software updates.
Select Keystone	Select the keystone calibration file. The Standard setting is the camera-specific keystone calibration for your lens (6 or 8 mm).

On-camera Blackbalance	On-camera Balance by on-sensor black pixels. This setting is relevant when an exposure value between 35% and 80% of the maximum exposure is used (this results in step-like artifacts in the black image). If the problem does not occur, set Never . If a trigger signal is used, enable Triggered-only (increases noise a little, but no more step-artefact occurs). If blackbalance is performed while not triggering but using it for triggered mode, activate Always (Blackbalance will be captures with the same settings as it is applied).
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5.7 CAPTURE TAB – BLACKBALANCE / WHITEBALANCE



Blackbalance:

1. Close the camera lens with the lens cap.
2. Start the recording via **Record Blackbalance**.
3. The calibration will be automatically applied, unless disabled

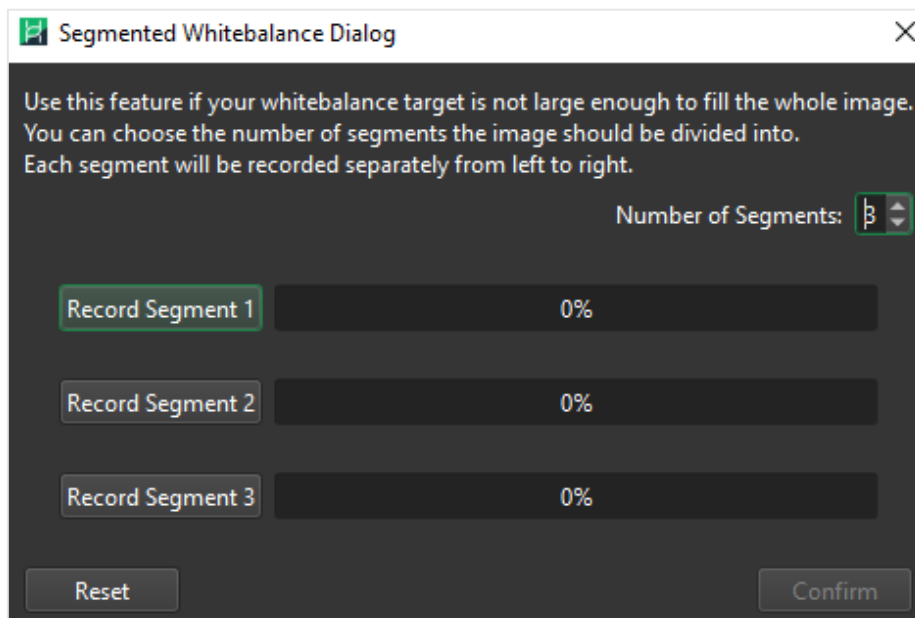
Important: Balance will not be saved when closing the software, but you can load the Blackbalance from the previous session by clicking **Load last Blackbalance** if it fits your selected mode (Further information in chapter 10.1).

Caution: As soon as a new framerate has been set, a new BlackBalance must also be recorded.

Whitebalance:

1. Place a white reference target under the camera which is suitable for your application.
2. Start the recording via **Record Whitebalance**.
3. The calibration will be automatically applied, unless disabled.
4. If the available white reference target is not big enough, select **Segmented Whitebalance**:
 - o Each segment is recorded in turn from **left to right** so that the white panel can be moved in the meantime.

- Adjust the number of required segments and press **Record Segment 1** to **Record Segment X**.
- **Confirm** the complete Whitebalance.



Important: Balance will not be saved when closing the software, but you can load the Whitebalance from the previous session by clicking **Load last Whitebalance** if it fits your selected mode.

5.8 CAPTURE TAB – FPS MONITORING

This section monitors the **FPS** (frames per second) during livestreaming.

- The FPS values of **Receive**, **Preprocessing** and **Processing** should correspond to the set **Framerate** in camera settings.
- The FPS value of **Display** should always be approx. 10 Hz.
- **Lost Frames** indicates problems either during the Processing on device (P), the Ethernet Bandwidth (E) or the User-Software (U). The ideal is to have a value of 0. When the **Lost Frames** value is above 0, please check your setup (also Pixel Packing settings) or contact the support team of HAIP Solutions GmbH.

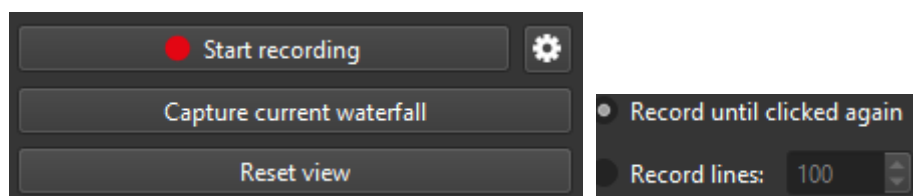
Positive example – no lost frames:

FPS	
Receive:	100.48
Preprocessing:	100.45
Processing:	101.47
Display:	9.95
Lost Frames:	0 (0 P / 0 E / 0 U)

Negative example – lost frames identified:

FPS	
Receive:	99.90
Preprocessing:	99.90
Processing:	99.05
Display:	9.34
Lost Frames:	33 (0 P / 33 E / 0 U)

5.9 CAPTURE TAB – DATA RECORDING



Start/Stop Recording:

- **Settings button:** Choose whether to record until pressing **Stop Recording** again (default setting) or select a specific number of lines to record. The recording will stop automatically afterward.
- **Start Recording:** the software starts recording the lines that are acquired by the camera.
- **Stop Recording:** the software stops the recording and directs automatically to the **Analysis Tab**.

Capture current waterfall:

- All lines, currently displayed in the **Waterfall window**, are captured. The resulting image will be automatically displayed in the **Analysis Tab**.

Reset view:

- Resets arrangement of Livestream windows.

6 BLACKSTUDIO - ANALYSIS TAB

The **Analysis tab** in the BlackStudio software can be used to visualize and then analyze previously captured hyperspectral data. The structure and functions of the **Analysis tab** are described below.

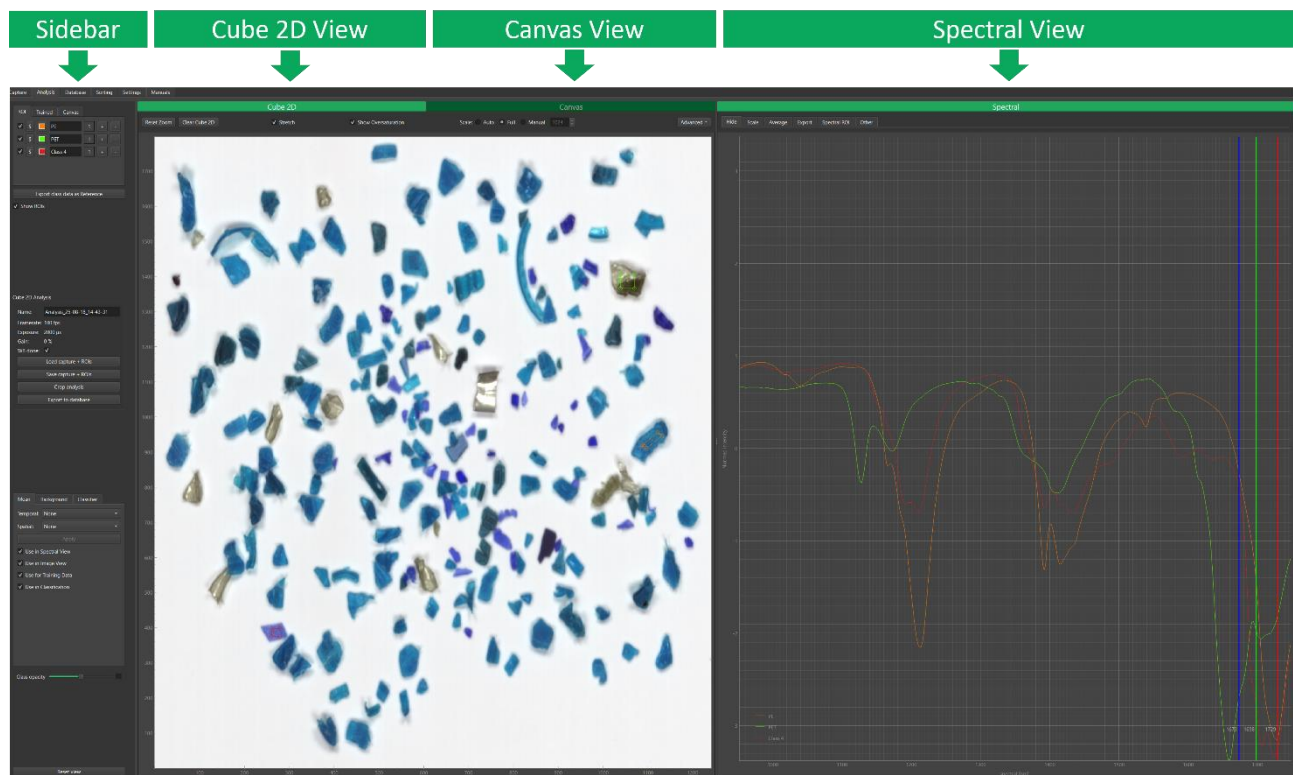
6.1 ANALYSIS TAB – OVERVIEW

The **Analysis tab** consists of four sections, which are explained in detail in the following sub-chapters:

- The **Sidebar**
- The **Cube 2D View**
- The **Canvas View**
- The **Spectral View**

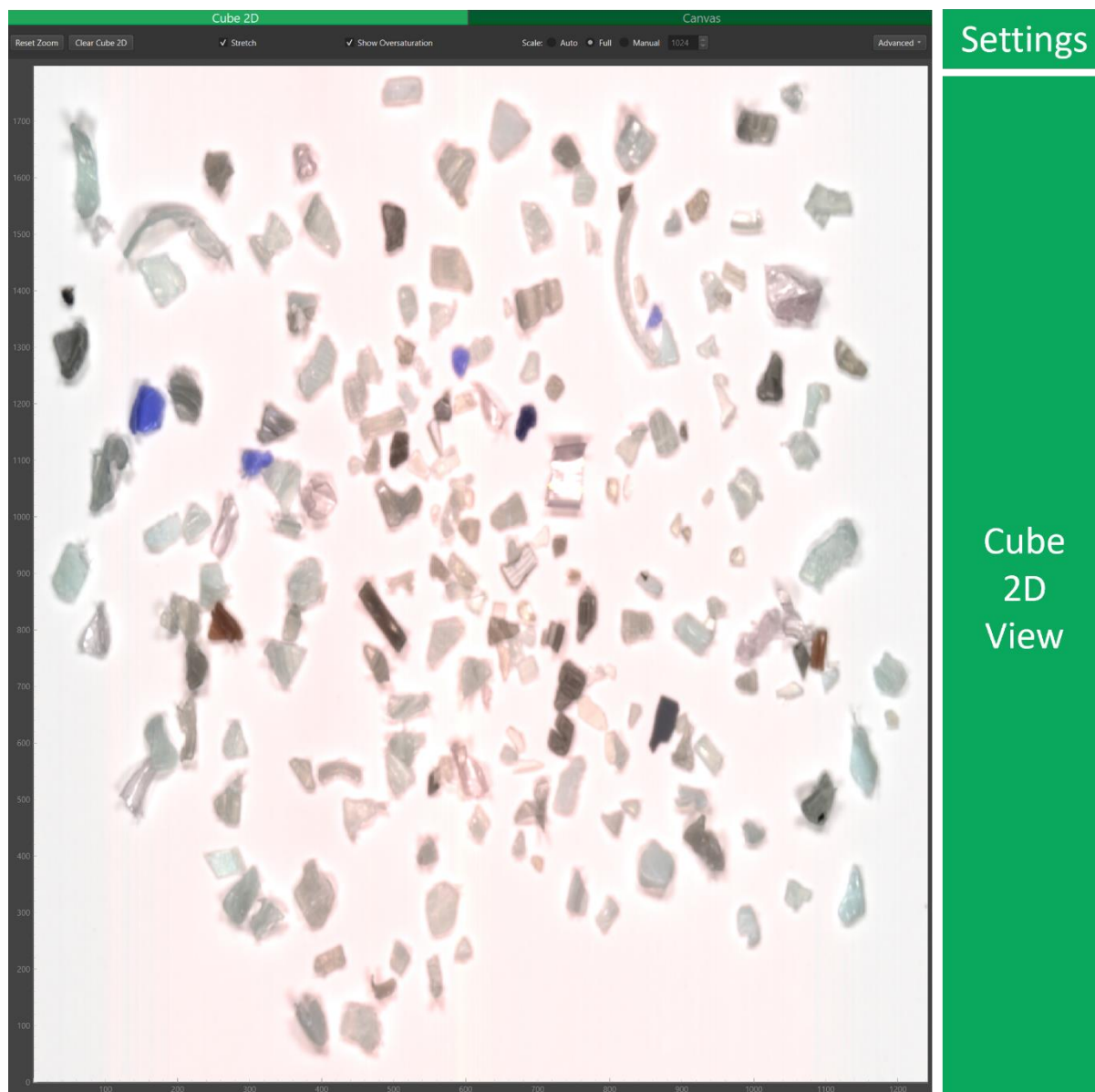
Identical to the Capture Tab, the three sections can be displayed variably:

- The three view windows, indicated by the green title bar, can be freely arranged by drag & drop or resized using the mouse pointer.
- Open the individual windows to full screen by double clicking on the dedicated green title bar.
- Via the **Reset View** button, the default arrangement of the windows can be regained.



6.2 ANALYSIS TAB – CUBE 2D VIEW

The screenshot below shows the Cube 2D View within the Analysis tab with an exemplary loaded hyperspectral data cube.

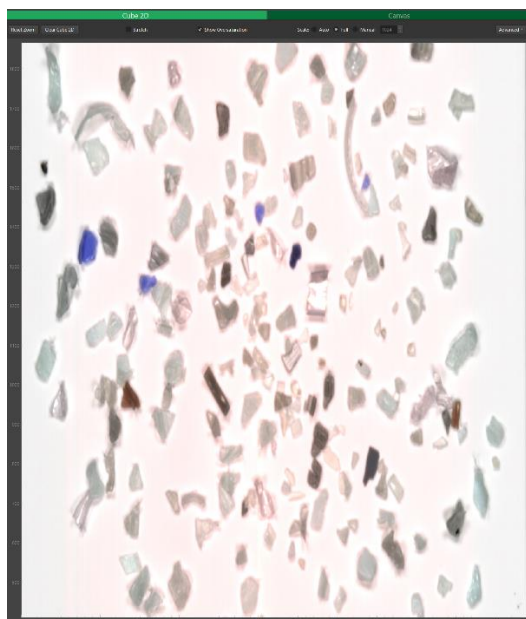


Settings:

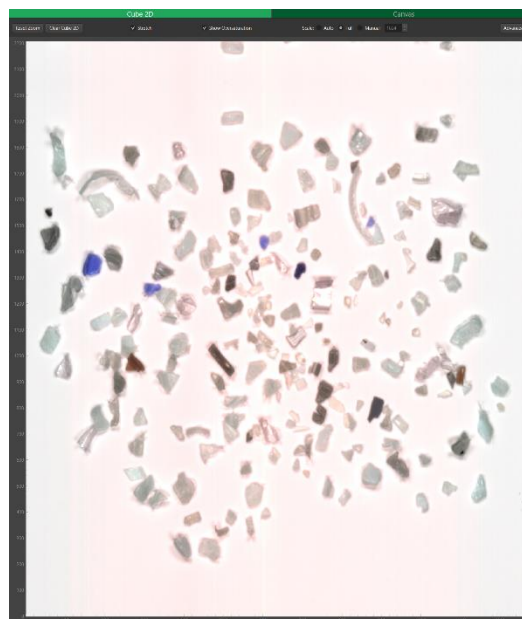
- **Reset Zoom:**
 - You can zoom in and out as required in the **Cube 2D View**. The full extent can be restored by pressing **Reset Zoom**.
- **Clear Cube 2D:**
 - Clears the **Cube 2D View** including hyperspectral cube and ROI markers.
 - A warning will appear before deletion: "Clearing Cube 2D View will reset all ROI markers saved on the Cube2D image and deletes the image! Make sure to save your ROIs and image beforehand".
- **Stretch:**

- Activated by default.
- When activated: The scale of the hyperspectral cube is adapted to the size of the 2D Cube display.
- When deactivated: the hyperspectral cube is displayed in its original extent. This extent depends on several settings, such as the **Framerate** and the speed of the conveyor belt.
- To examine the entire data cube, you can scroll around (up/down/left/right).

Stretch deactivated:



Stretch activated



• Show Oversaturation:

- Activated by default.
- When activated: Oversaturated pixels are displayed in red in the **Cube 2D View**. Oversaturated pixels will not be included in the mean spectra of ROI markers.
- When deactivated: Oversaturated pixels will not be highlighted. The oversaturated pixels will be included in the calculation of mean spectra of ROI markers.

• Scale:

- **Auto:** The intensity is scaled automatically. The automatic scale aligns the highest measured intensity to the maximum gray scale.
- **Full:** The full scale varies depending on the radiometric resolution of the camera.
- **Manual:** Set the maximum intensity value manually.

• Export:

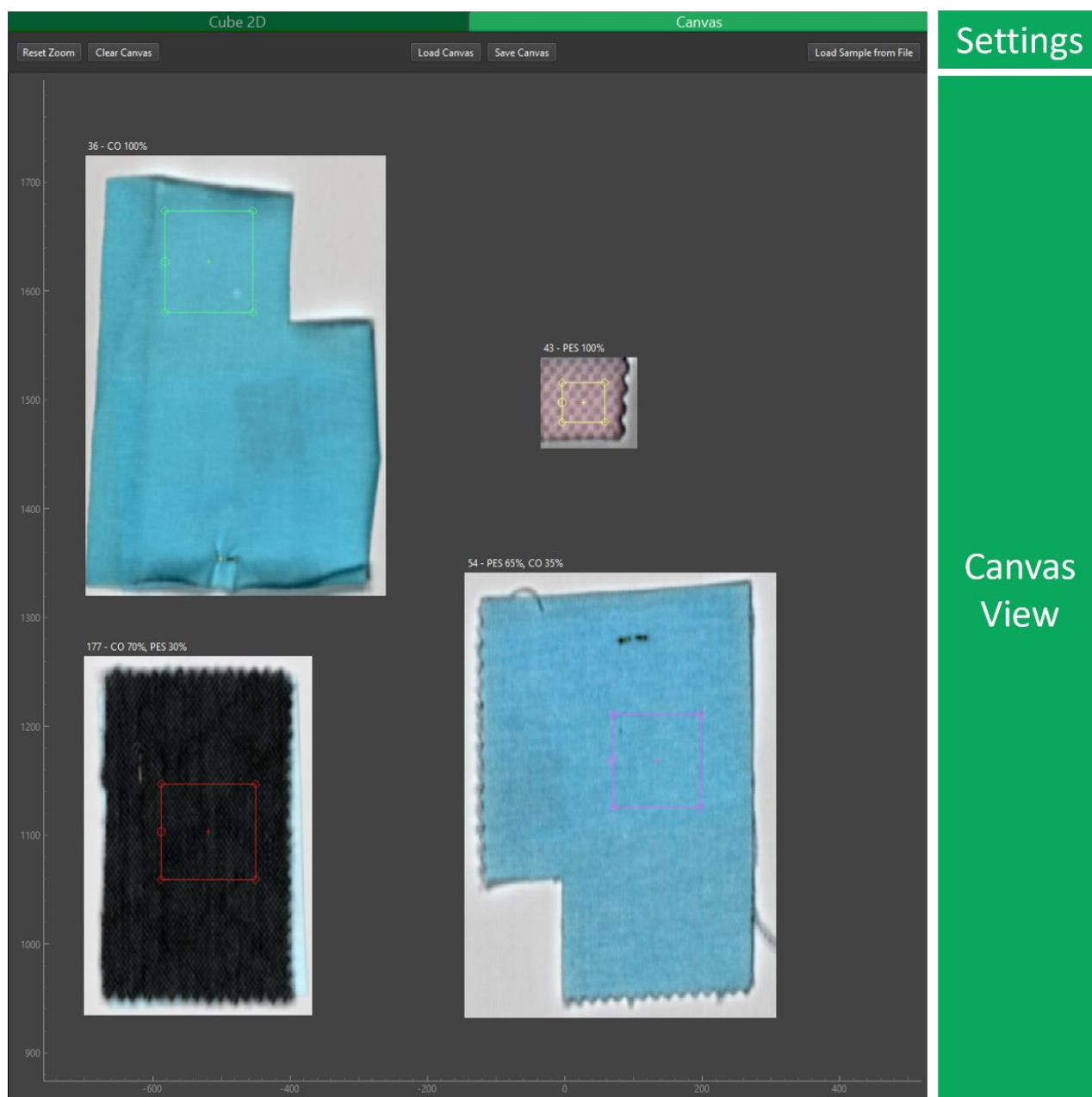
- **Export Cube 2D Window as .png:** Export the currently displayed **Cube 2D View** as *.png file (including ROI marker, side axes).
- **Export HSI data as False-Color .png:** Export the entire hyperspectral data cube in the currently set false color representation as *.png image. ROI markers are not exported, and the stretch is not taken into account. This allows a higher image resolution to be achieved.

Cube 2D View:

- **Color scheme:**
 - The color scheme in the **Cube 2D View** can be manipulated by the three sliders in red, green and blue in the **Spectral View**.
- **Zoom:**
 - The **Cube 2 D View** can be zoomed in and out by scrolling. The aspect ratio of the pixels remains unchanged.
 - Holding the right mouse button and moving the cursor to the right causes the pixels to be enlarged in X direction.
 - Holding the right mouse button and moving the cursor upwards causes the pixels to be enlarged in Y direction.
 - To reset a zoom, please press **Reset Zoom**.
 - Press and hold CTRL and scroll the mouse wheel to scroll vertically without zooming.

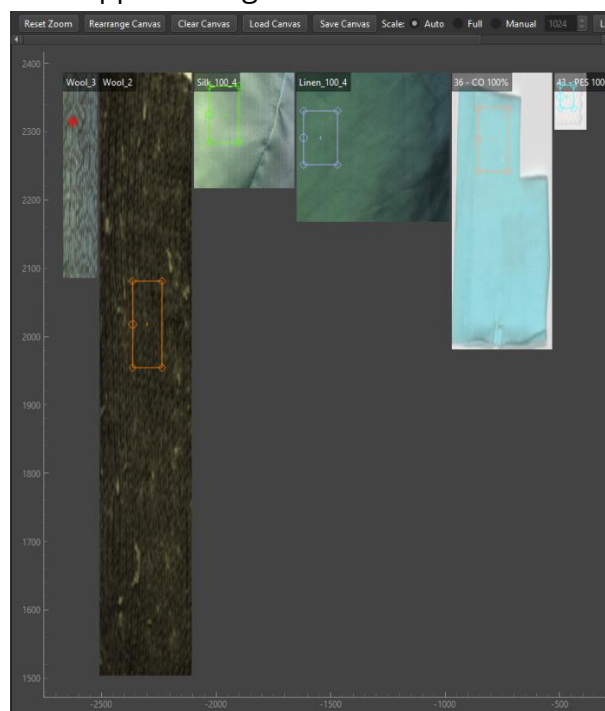
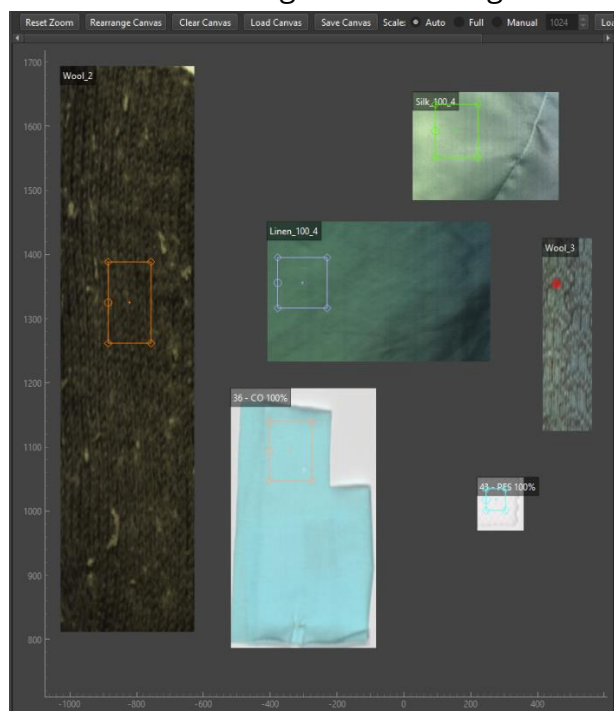
6.3 ANALYSIS TAB – CANVAS VIEW

The **Canvas View** enables the visualization, analysis and classification of image snippets that are stored in the **Database**. All snippets and the associated ROIs can be freely arranged in **Canvas View**. Snippets and ROI markers can be moved together or separately. New ROI markers can also be created.



Settings:

- **Reset Zoom:**
 - You can zoom in and out as required in the **Canvas View**. The full extent can be restored by pressing **Reset Zoom**.
- **Rearrange Canvas:**
 - Pressing the button aligns the loaded snippets along the current zoom level.



- **Clear Canvas:**
 - Clears the Canvas View including hyperspectral cube and ROIs
 - A warning will appear before deletion: "Clearing Canvas view will reset all ROI markers saved on the Canvas! Make sure to save your ROIs beforehand."
- **Load Canvas:**
 - Load a previously stored full **Canvas View** including all the snippets it contains.
- **Save Canvas:**
 - Save the currently displayed **Canvas View**, including its contained data (*.hdr & *.img).
- **Scale:**
 - **Auto:** The intensity is scaled automatically. The automatic scale aligns the highest measured intensity to the maximum gray scale.
 - **Full:** The full scale varies depending on the radiometric resolution of the camera.
 - **Manual:** Set the maximum intensity value manually.

- **Load Sample from File:**
 - Load a previously created data cube into the canvas view. Accordingly, this does not have to be stored in the database beforehand. Caution: Large files may take some time to load.
- **Export:**
 - **Export Canvas Window as .png:** Export the currently displayed the current Canvas view as *.png file (including ROI marker and side axes).
 - **Export HSI data as False-Color .png:** Export the entire hyperspectral data cube in the currently set false color representation as *.png image. ROI markers are not exported, and the stretch is not taken into account. This allows a higher image resolution to be achieved.

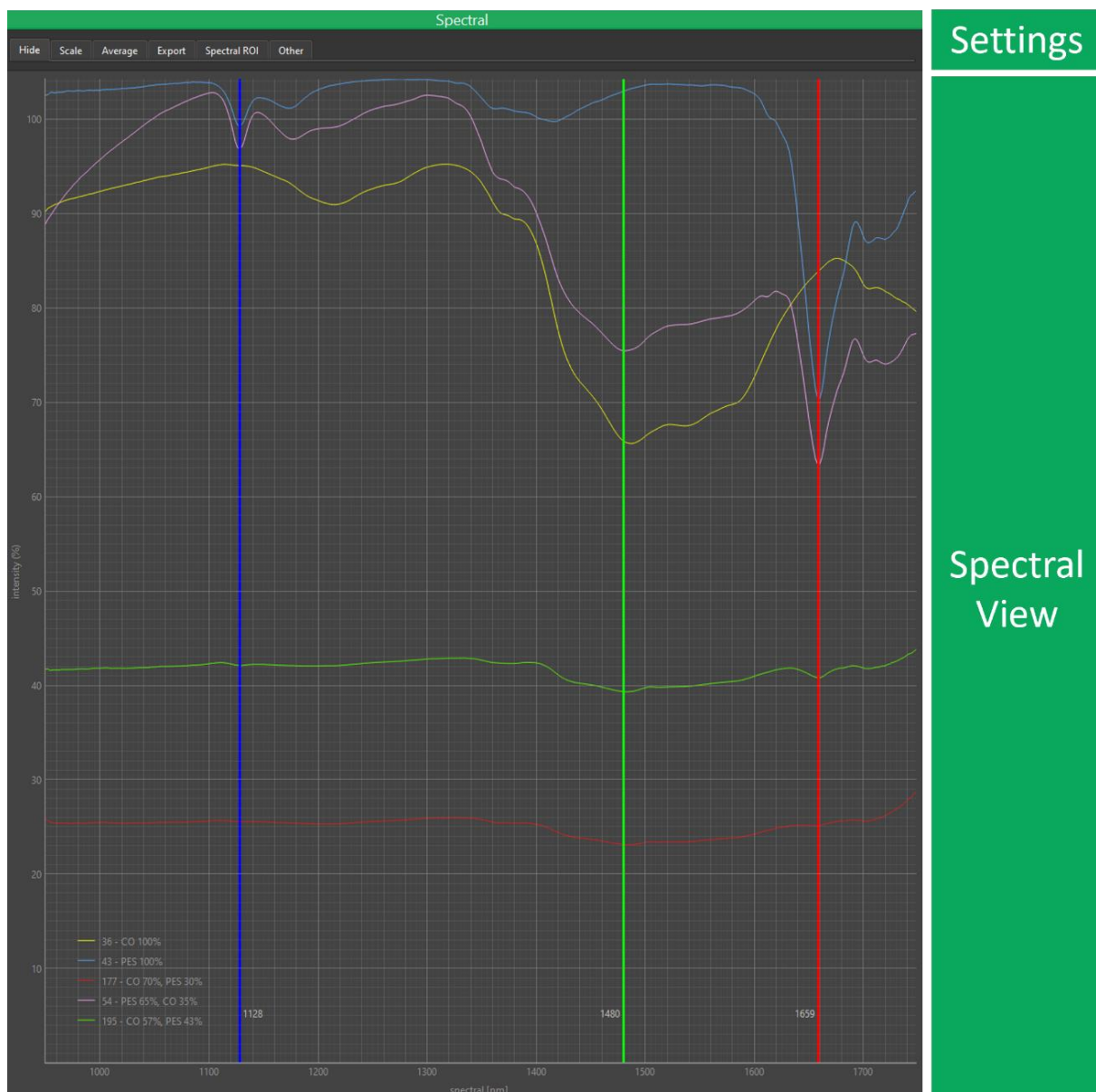
Canvas View:

- **Color scheme:**
 - The color scheme in the **Canvas View** can be manipulated by the three sliders in red, green and blue in the **Spectral View**.
- **Zoom:**
 - The **Canvas View** can be zoomed in and out by scrolling. The aspect ratio of the pixels remains unchanged.
 - Holding the right mouse button and moving the cursor to the right causes the pixels to be enlarged in X direction.
 - Holding the right mouse button and moving the cursor upwards causes the pixels to be enlarged in Y direction.
 - To reset a zoom, please press **Reset Zoom**.
 - Press and hold **CTRL and scroll the mouse** wheel to scroll vertically without zooming.

6.4 ANALYSIS TAB - SPECTRAL VIEW

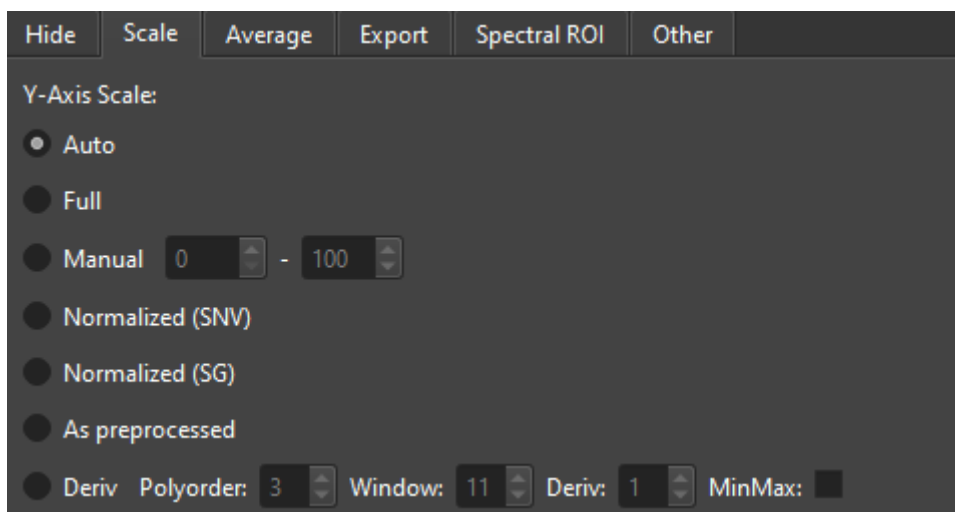
In the **Spectral View**, the spectral intensity curves of pre-defined ROI-classes are displayed.

- All spectral curves of all ROIs are displayed, from the **Cube 2D View** as well as from the **Canvas View**. In the settings bar in the **ROI window**, the spectral curves of existing ROIs can be displayed or hidden by setting a check mark. For more information on ROI window, see chapter 6.5.1.
- By sliding the red, green and blue slider, adjust the color presentation of the currently opened **Cube 2D View** or **Canvas View**.
- By scrolling or holding the right mouse button and moving the mouse pointer in the **Spectral View**, it is possible to zoom in the X direction so that a desired wavelength range is enlarged.



Settings:

- **Scale:**



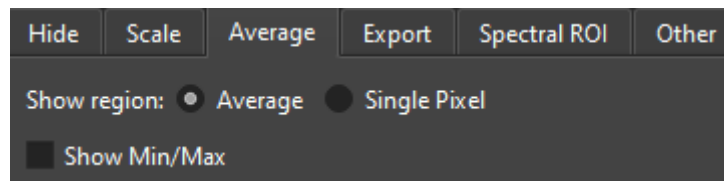
Adjust the **Y-axis scale** according to your needs. The intensity level on the Y-axis varies depending on which samples are currently loaded. If the data is already white referenced, the intensity level is given in percentage %, otherwise in the absolute values in digital numbers (DN).

- **Auto:** Automatic adjustment to intensity levels of currently selected ROIs
- **Full:** Full stretch (either % or DN)
- **Manual:** Manual selection of the Y-axis range to be displayed (0–100 % or 0 – max. DN)
- **Normalized (SNV):** SNV (Standard Normal Variate) is calculated by subtracting each spectrum by its own mean and dividing it by its own standard deviation. Therefore, after SNV, each spectrum will have a mean of 0 and a standard deviation of 1.
- **Normalized (SG):** Normalization based on (Stokman and Gevers) Norm using this formular:

$$R_{SG}(\lambda) = \frac{R(\lambda)}{\sum_{i=1}^N R(\lambda_i)} - \min_{j \in [1, N]} \frac{R(\lambda_j)}{\sum_{i=1}^N R(\lambda_i)}$$

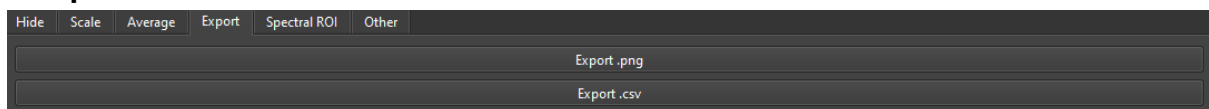
- **As preprocessed:** Select the data display as it was set during the preprocessing of the classification (more information see chapter 6.5.3).
- **Deriv:** Displays the intensity spectra in the first, second or third **derivative**. Possibility of selecting the polyorder and the window. **Savitzky-Golay smoothing is always applied.** If the zero derivative is set, the original data is smoothed accordingly. Min-Max normalization by activating **MinMax** checkbox.

- **Average:**



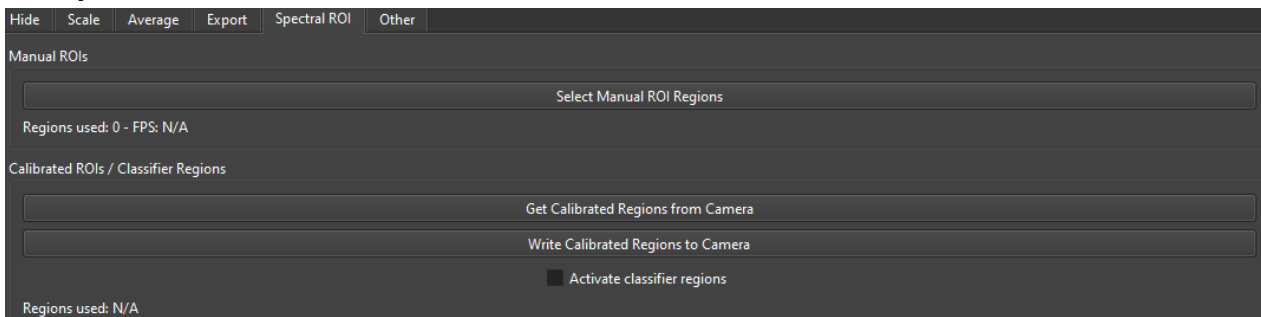
- **Show region – Average:** Provides the intensity values of the entire selected ROI-class as average.
- **Show region – Single Pixel:** Provides the intensity values of the selected ROI-class from the center pixel of the respective ROI (center pixel is marked in ROI).
- **Show Min/Max:** Activate the checkbox to display the value range of the entire ROI from minimum to maximum.

- **Export:**



- **Export .png:** Export the displayed Spectral View as an image (*.png).
- **Export .csv:** Export the intensity values of the displayed spectral curves in *.csv file format.

- **Spectral ROI:**



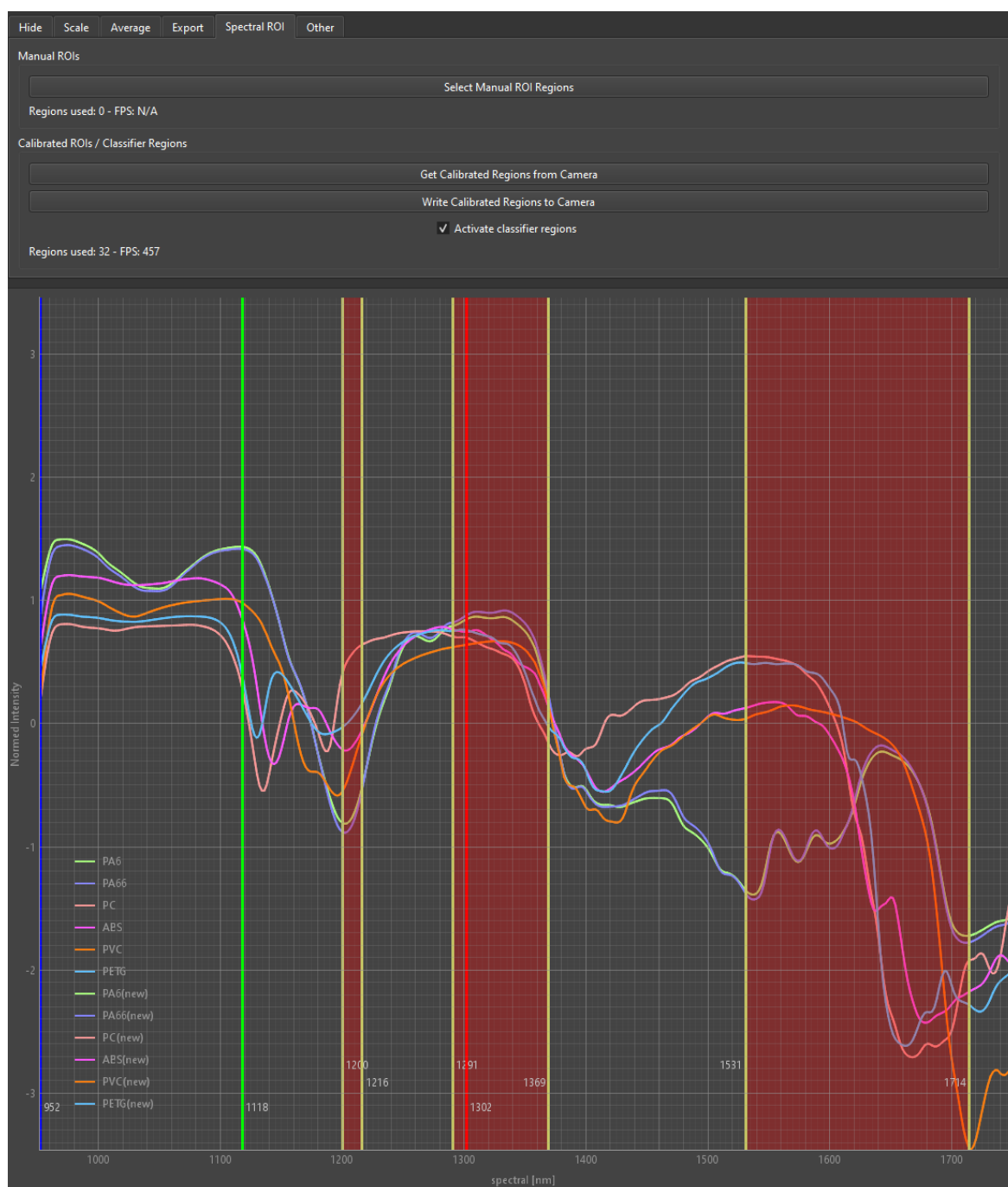
Please note that this function is only available for cameras of the BlackIndustry SWIR series! More information on how to apply Spectral Regions (ROIs) can be found in chapter 10.4. This chapter only describes the basic settings.

In order to use the **Multiple-ROI functionality** (Readout Mode of camera, see chapter 10.4), the desired **Spectral Regions** have to be defined first in **Calibrated ROIs / Classifier Regions**:

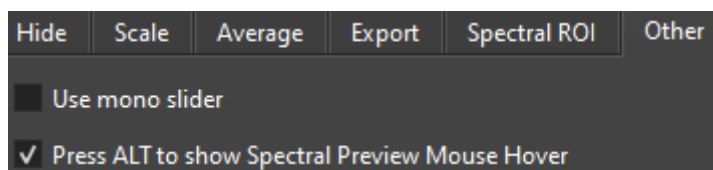
- Calibrated spectral ROIs can be selected and displayed by activating the checkbox **Activate classifier regions** or by pressing **Get Calibrated Regions from Camera**.
- By double-clicking in the **Spectral View**, Spectral Regions can be created, displayed as red bars. The size of the regions can be changed by dragging the sliders. Right-click to delete a region. A maximum of 8 regions can be created. The number of selected sensor regions within the Spectral Regions is indicated under **Regions used**.

In addition, the frame rate in fps that can be achieved with the selected number of regions is calculated at the same time.

- Press **Write Calibrated Regions to Camera** to transfer the Spectral Regions to the camera. The Spectral Regions are stored on the camera until new Regions are set. Even rebooting the camera does not delete the Spectral Regions
- To access the Spectral Regions stored on the camera, press the **Get Calibrated Regions from Camera** button.
- For more information on Spectral Regions / Classifier regions, see chapter 11.4 & 11.5.3.



- **Other**



- **Use mono slider:** Activate checkbox to change the color coding of the **Cube 2D View** or **Canvas View** to grayscale. Move the slider to select the desired wavelength to be displayed.
- **Press ALT to show Spectral Preview Mouse Hover:** Activate the checkbox and press the ALT key on the keyboard. If the cursor is now moved over the samples in the **Cube 2D View** or in the **Canvas View**, a preview of an intensity curve of the corresponding pixel is displayed in the **Spectral View**.

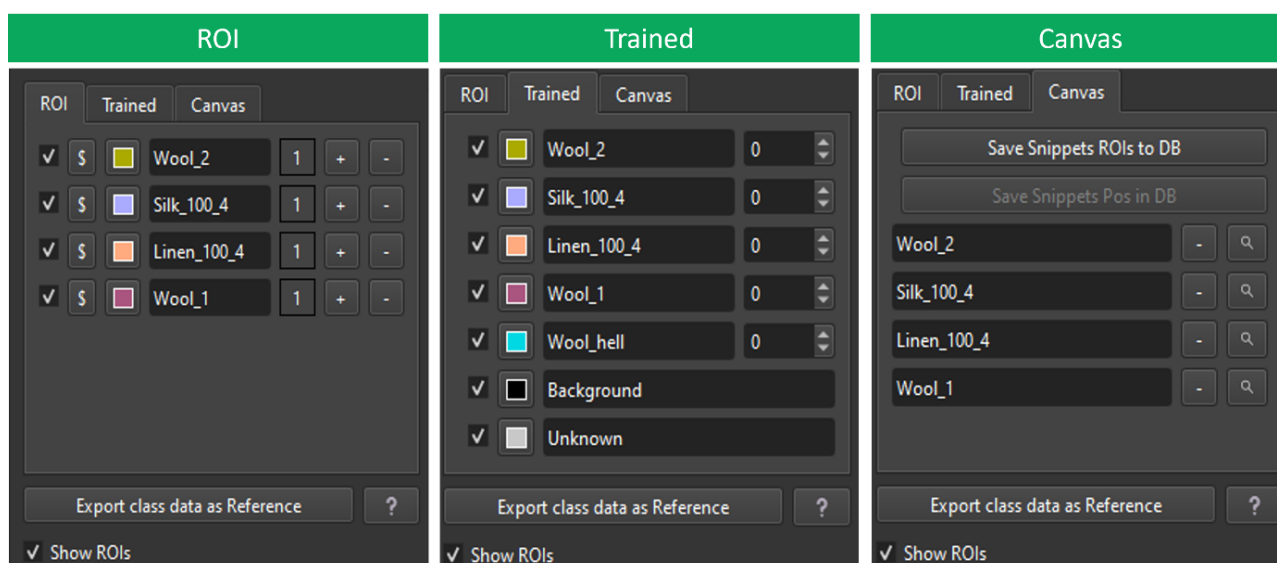
6.5 ANALYSIS TAB – SIDEBAR

The **sidebar** in the **Analysis Tab** is divided into three sub-categories:

- The **Data and ROI** panel
- The **Cube 2D Analysis** panel
- The **Mean/Background/Classifier** panel

6.5.1 SIDEBAR – DATA AND ROI

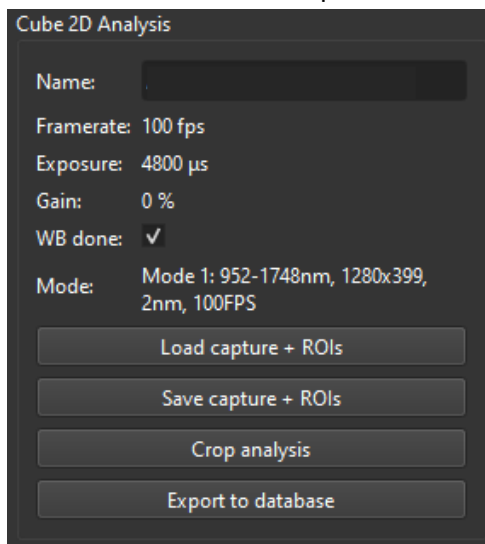
The **Data and ROI** segment in the upper third of the settings section provides information on which ROIs are currently displayed in the **Cube 2D View** and in the **Canvas View**. You can switch between three tabs:



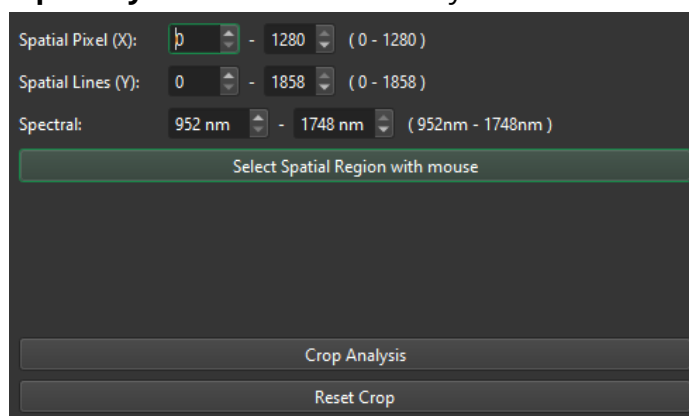
- **ROI:**
 - All newly created or imported ROIs are displayed in the **ROI** section.
 - Assign a color and a meaningful name to each ROI.
 - The **\$** button can be used to select predefined classes including coloring if required (The classes can be customized according to requirements under /config/sample-table.csv in the program directory). Alternatively, previously created materials from the database can also be assigned.
 - Add additional ROIs of the same class or delete them using the **+** and **-** symbol.
 - The current number of ROIs displayed for each class is indicated by the number next to it. If you add reference data to the ROI, the background gets colored green. (More information see **Export / Load class data as reference**).
- **Trained:**
 - When a classifier was trained based on ROIs, these ROIs are displayed in the **Trained tab** with color coding and name.
 - The Background is displayed in black by default and Unknown, unclassifiable areas are gray by default.
 - The number behind each trained ROI indicates the **threshold** for the 'Unknown' class (default 1000 for LDA, 0 for SVM). If the value is reduced, pixels that are closest to the class but further away than the set threshold are classified as Unknown. The threshold is specified in a Euclidean distance, but in a transformed space, so it has no real equivalent.
- **Canvas:**
 - This tab lists all ROIs and snippets that can be found in the canvas view. By pressing the **-** button behind it, the ROIs can be removed together with the snippets. Click the **lens icon** to zoom in on the corresponding snippet.
 - **Save Snippets ROIs to DB:** New ROIs can be created in the **Canvas View** and existing ones can be changed. If the changes are to be saved in the database, press the button. If the snippets are then reloaded from the database, they are displayed as they were when the button was pressed.
 - **Save Snippets Pos in DB:** If a snippet group is loaded into the Canvas without any additional individual snippets, you can use **Save Snippets Pos in DB** to save the currently displayed position and arrangement of the snippets. This feature is only available for snippet groups.
- **Export class data as Reference:**
 - When clicked, the displayed ROIs from the **ROI** tab are saved as *.hpr file in a desired storage location. For advanced management of reference data, you might want to use the database.
- **Show ROIs:** To hide all ROIs, uncheck the box next to **show ROIs**.

6.5.2 SIDEBAR – CUBE 2D ANALYSIS

In this section on the left side in the **Analysis tab**, you can define a project name for the upcoming analysis. The capture settings for the currently displayed image are displayed (**Framerate, Exposure, Gain, Mode**). If a white balance was performed, **WB done** will be checked.



- **Load capture + ROIs:** Load previously captured images including ROI-classes into the **Cube 2D View**.
- **Save capture + ROIs:** Save the current hyperspectral image + ROI-classes (ROI-classes are written into the .hdr file of each ENVI format hyperspectral data cube).
- **Crop analysis:** Crop the region of analysis either in the spatial dimension in x- or y- direction or in the spectral dimension by cropping the wavelength range.
 - Choose **Select Spatial Region with mouse** to select an area on the **Cube 2D View** with the mouse which is to be cut out.
 - Press **Crop Analysis** to confirm the entry.



- **Export to database:** (RECOMMENDED FOR BEST MANAGEMENT OF SAMPLES) Export a selected area from **Cube 2D View** with all associated information to the **database**. These "**Snippets**" can later be loaded together in the **Canvas View**. (For more information, see chapter 11.2 & 11.3).

6.5.3 SIDEBAR – MEAN/BACKGROUND/CLASSIFIER

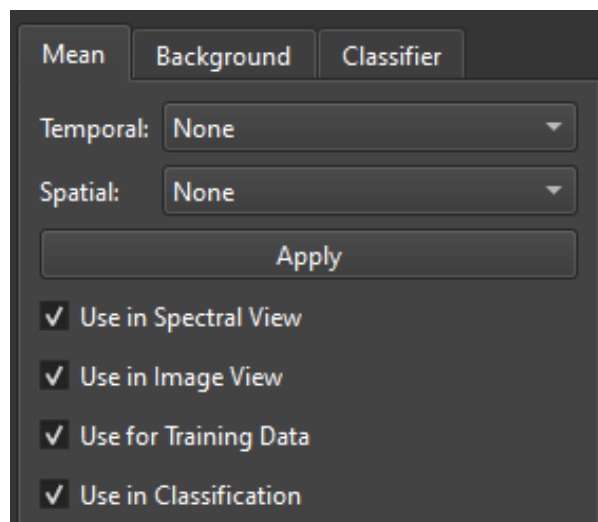
This section is divided into three tabs:

- **Mean**
- **Background**
- **Classifier**

Mean:

In this tab, a mean filter can be applied to the data.

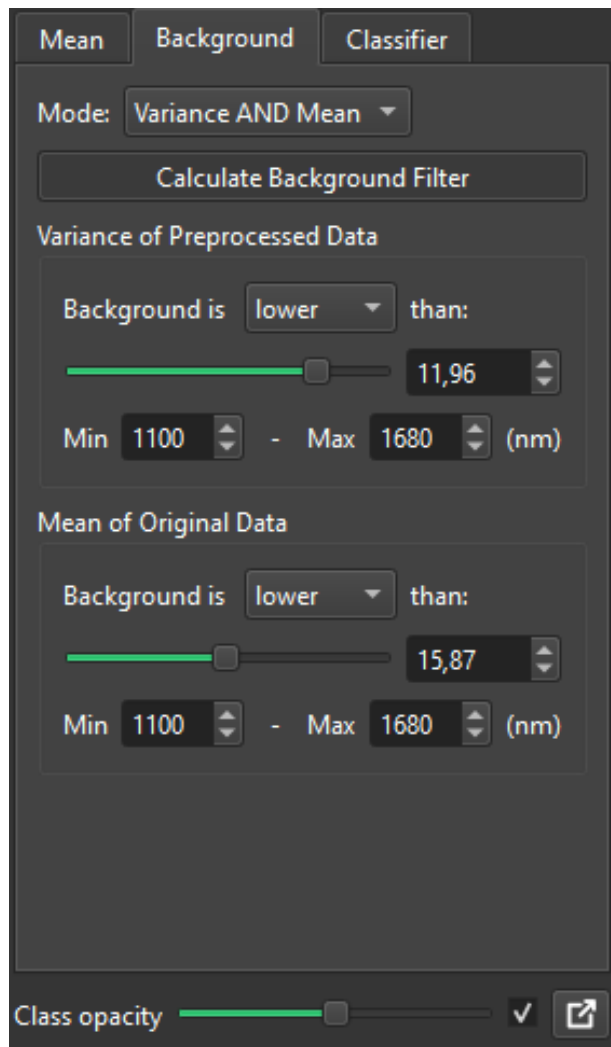
- **Temporal:** By setting a mean filter in the temporal direction, pixel values in a defined neighborhood (3, 5, 7, 9 pixels) are replaced by their mean values. This leads to an effective noise reduction.
- **Spatial:** By setting a mean filter in the spatial direction, pixel values in a defined neighborhood (3, 5, 7, 9 pixels) are replaced by their mean values. This leads to an effective noise reduction.
- **Use in Spectral View:** Enable the checkbox if the mean filter is to be applied in the Spectral View window.
- **Use in Image View:** Enable the checkbox if the mean filter is to be applied in the Cube 2D and Canvas View window.
- **Use for Training Data:** Enable the checkbox if the mean filter is to be used for training a classifier.
- **Use in Classification:** Enable the checkbox to apply mean filter in classification.



Background (BG):

In this tab, a Background filter can be created and adjusted. The background filter filters out the background of the image so that it is not included in the later classification (see also chapter 11.4.2).

- **Mode:** None / Variance / Mean / Variance AND Mean / Variance OR Mean / Variance x Mean
- **Calculate Background Filter:** Press to calculate the background filter with the currently set variance and /or mean values. Depending on the image size, it may take a few seconds. The background filter is displayed in the **Cube 2D View** and **Canvas View**. The black areas are marked as background. The light areas should mark the objects of interest. Adjust the opacity level using the slider below.
- **Variance of Preprocessed Data:** The background filter can now be modified in real time. Depending on the Mode, change the settings of Variance of Preprocessed Data.
- **Mean of Original Data:** The background filter can now be modified in real time. Depending on the Mode, change the settings of Mean of Original Data.



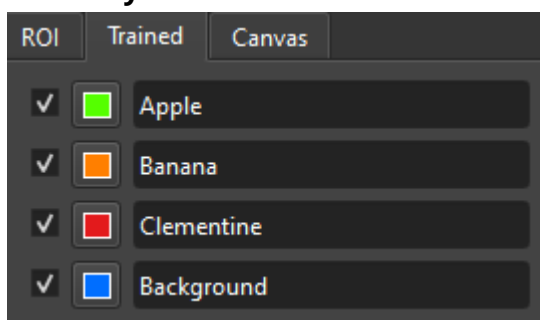
The screenshot shows a software interface for background filtering. At the top, there are three tabs: 'Mean', 'Background' (which is selected), and 'Classifier'. Below the tabs, the 'Mode' is set to 'Variance AND Mean'. A button labeled 'Calculate Background Filter' is present. The interface is divided into two main sections: 'Variance of Preprocessed Data' and 'Mean of Original Data'. Each section contains a dropdown menu set to 'lower', a slider, and a numerical value. For 'Variance of Preprocessed Data', the value is 11,96. For 'Mean of Original Data', the value is 15,87. Both sections also have 'Min' and 'Max' input fields with values 1100 and 1680 respectively, followed by '(nm)'. At the bottom, there is a 'Class opacity' slider and a checkmark icon.

Classifier:

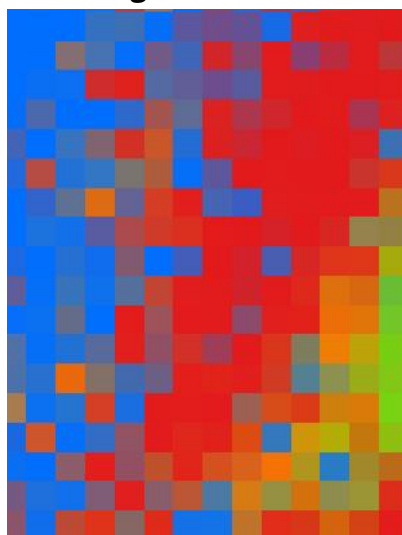
A classifier can be trained and applied using the **Classifier** menu. How to create an ideal classifier is explained in chapter 11.4.

- **Name:** Assigning a name to the classifier makes it easier to identify it later when the classifier is saved.
- **Exclude BG from Training Data:** Even if the defined ROI markers include background, this area is excluded from the classifier training. This allows the ROI markers to be drawn larger, across several fragments (e.g., plastic flakes, rice grains, etc.) of a class.
- **Use BG Filter for Classification:** Activate the checkbox to include the Background Filter in the Classification.
- **Visualize Unknown Class (SVM+):** Only available for SVM+ Classifier. Histogram shows distance values for all pixels in a class, displayed in the color of the class. The threshold for the Unknown class is visualized here and can be adjusted manually by moving the slider.
- **Disable Unknown Class:** Activated by default. Uncheck the checkbox to include the **Unknown** class in the classification. The threshold for the Unknown class can be adjusted for each class in **Trained**.
- **Probability Class Color Mixing:** For classifiers only. When enabled, individual pixels are colored during classification based on their probability of belonging to the different classes. Each pixel color is created by blending the original class colors to reflect the probabilities of belonging to a class. The class colors are therefore continuous rather than discrete.

Previously trained classes:

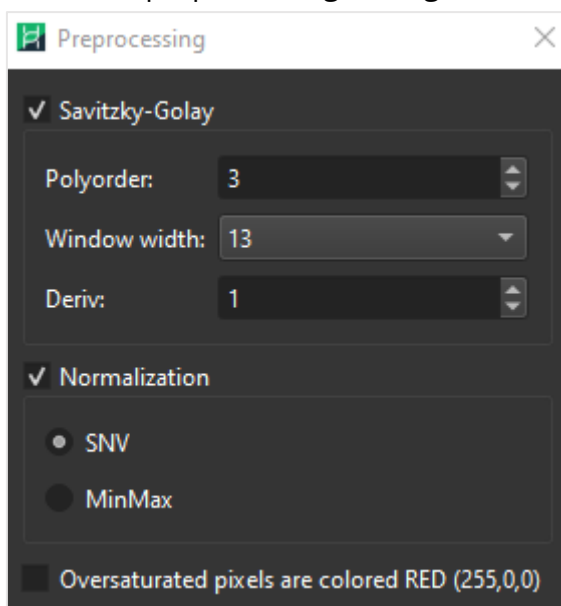


Color Mixing classification result:



Model:

- Choose between **Classifier**, **Regression** or **Visualization**:
 - **Classifier:**
 - **Classifier Model:** LDA / SVM / SVM+ / Logistic / PLS-DA (more information see chapter 11.4.3).
 - **Regression:**
 - **Regression Model:** SVR (more information see chapter 11.4.4)
 - **Visualization:**
 - **Visualization Model:** PCA (more information see chapter 11.4.5)
- **Configure Preprocessing:** When training a classifier, data preprocessing can be included. **Caution: As soon as the preprocessing settings are changed, a new background filter must be calculated.**
 - Enable/disable **Savitzky-Golay** Smoothing
 - Select the desired **derivative** (Deriv.)
 - Enable/disable **Normalization** (**SNV** or **MinMax**, see chapter 6.4)
 - Select whether **oversaturated pixels** should be displayed in red in the classification results (Note: None of the other classes should be assigned the color red).
 - Shown below: the default preprocessing settings



- **Train classifier:** Press as soon as all relevant settings have been made, and the classifier is to be trained. This may take a few seconds.
- **Test classifier:** Press to apply the currently loaded classifier to the data in the Cube 2D View or Canvas View.

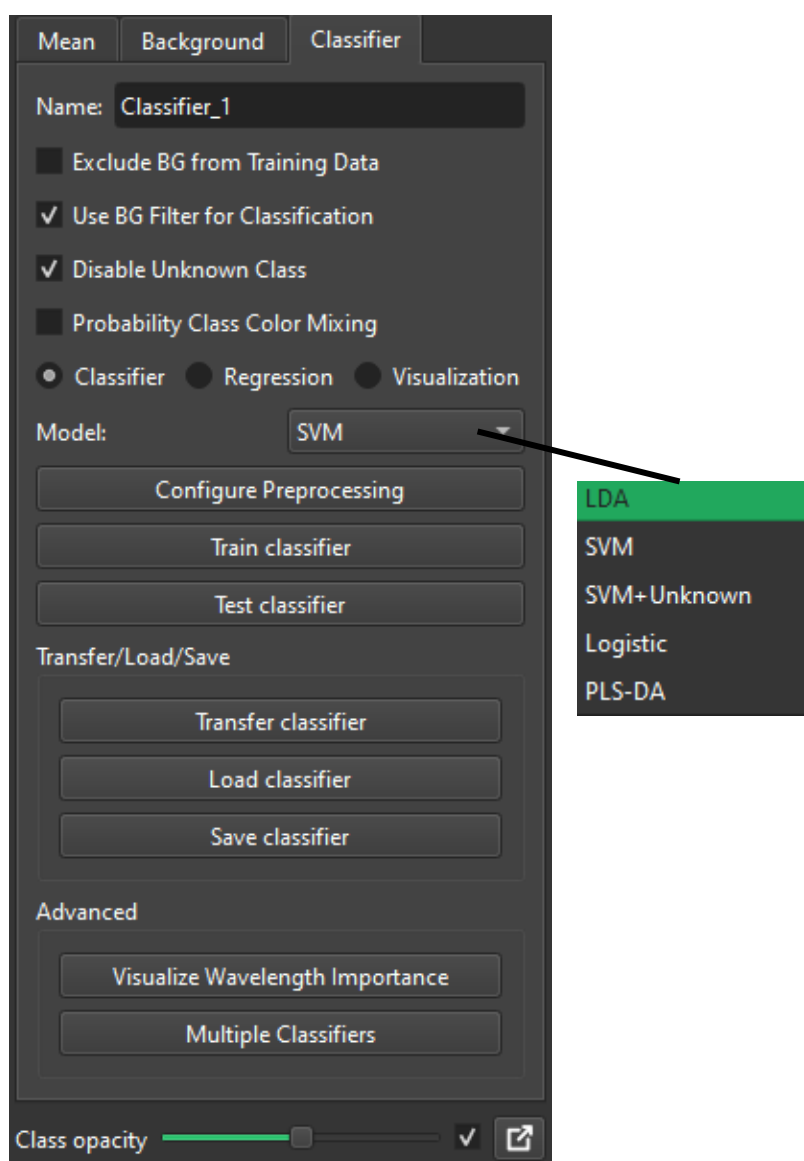
Transfer/Load/Save:

- **Transfer classifier:** For BlackIndustry SWIR cameras with firmware version 1.0.13 and later only. Once a classifier was created, it can be transferred to the currently connected camera. More information on On-camera classifier handling, see chapter 11.5.
- **Load classifier:** Load a previously created and stored classifier.

- **Save classifier:** Save the currently loaded classifier so that it can be applied to a different dataset.

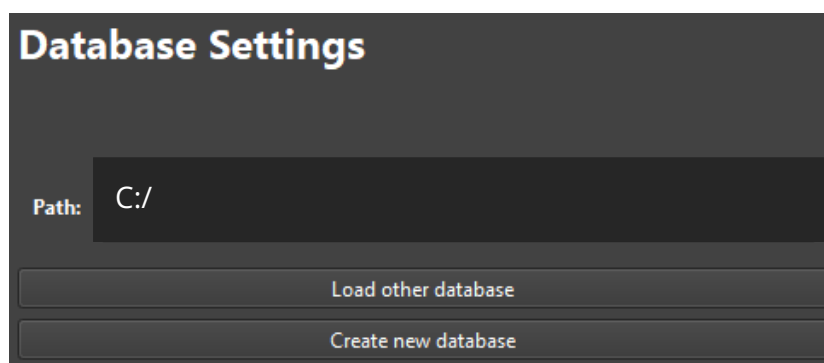
Advanced:

- **Visualize Wavelength Importance:** By pressing, this tool automatically calculates which wavelength regions are particularly important for successful classification. A new window opens showing the results. For more information, see chapter 11.4.6.
- **Multiple Classifiers:** Option to create multiple classifiers of the same type and run them simultaneously on the camera. For more information, see chapter 11.4.7.
- **Class opacity:** Slider for setting the opacity of the classification display. Uncheck the box on the right to hide the classification results.



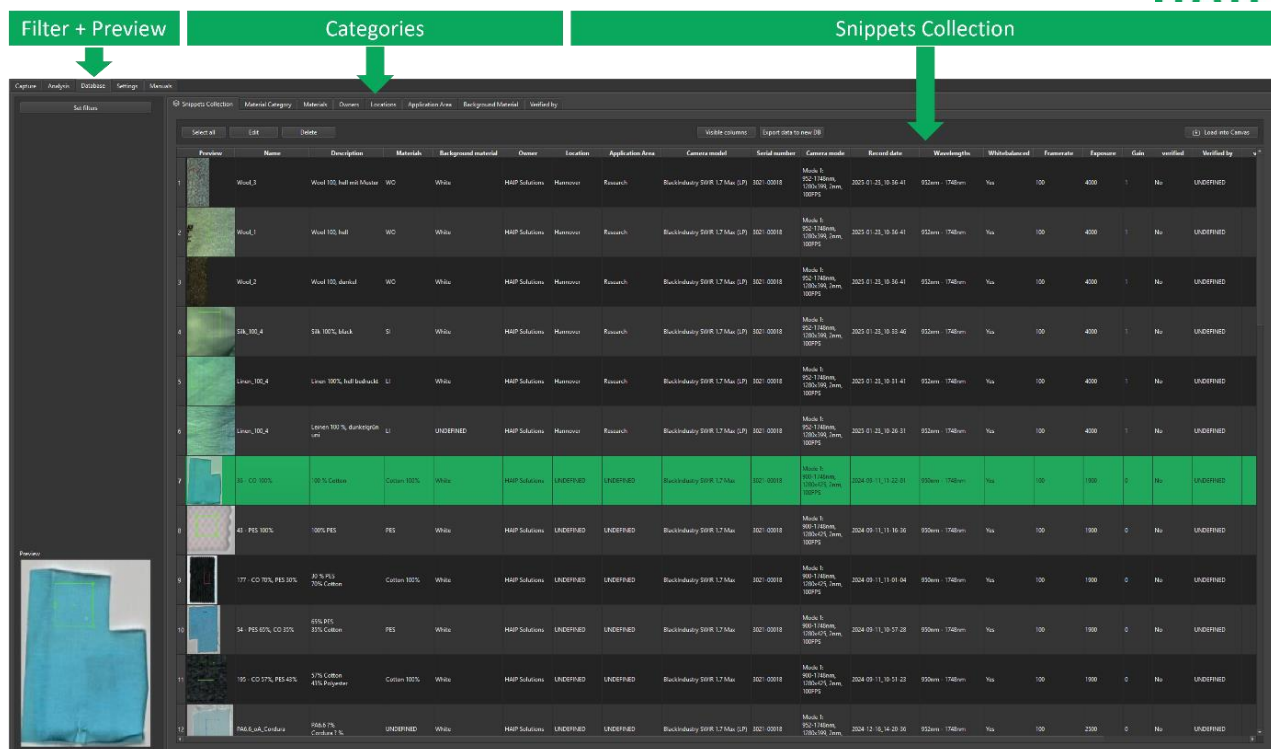
7 BLACKSTUDIO – DATABASE

The **BlackStudio Database** enables storage, sorting and filtering of collected data, the so-called **Snippets**. This chapter introduces the structure and main features of the database. Step-by-step instructions for creating and working with snippets can be found in chapter 11.2 and 11.3. When the BlackStudio software is started for the first time, you will be asked for the storage location of the database. If this is not subsequently changed, it remains unchanged. In the **Settings tab** of the software in the **Database Settings**, the storage path can be checked at any time, another database can be loaded, or a new one can be created. As the database is stored externally, it can also be copied and thus backed up.



7.1 DATABASE – OVERVIEW

The database consists of the **Snippets Collection** and seven tabs for each database category. On the left-hand side there is a filter option and the preview area of the snippets.



7.2 DATABASE – SNIPPETS COLLECTION

A **snippet** is a previously defined section of a hyperspectral image cube that is to be saved in the database. Further information on creating snippets can be found in chapter 8.2.

The following information is stored in the **Snippets Collection** for each snippet:

INFORMATION ON SNIPPET	ADDED MANUALLY / AUTOMATICALLY?
ID number	Auto
Preview image	Manual
Name	Manual (via Category)
Description	Manual (via Category)
Materials	Manual (via Category)
Background material	Manual (via Category)
Owner	Manual (via Category)
Location	Manual (via Category)
Application Area	Manual (via Category)
Camera model	Auto
Serial number	Auto
Camera mode	Auto
Record date	Auto

Wavelengths	Auto
Whitebalanced	Auto
Framerate	Auto
Exposure	Auto
Gain	Auto
Verified	Manual
Verified by	Manual (via Category)
Verification time	Auto
Change time	Auto
Creation time	Auto

The following actions can be executed in the **Snippets Collection**:

- **Select all:** Press to select all snippets in the database. Selected snippets will be marked in green.
- **Edit:** Select one snippet to edit its information.
- **Delete:** Select at least one snippet to delete them.
- **Visible columns:** Press to select which information columns will be shown in the Snippets Collection. **Tip:** Deactivating the preview image reduces the size of the rows.
- **Export data to new Database (DB):** Select at least one snippet and press the button to export the data to a new database folder.
- **Add to Snippet Group:** Select at least one snippet to be imported into a Snippet Group.
- **Remove from Snippet Group:** Select at least one snippet to be removed from a Snippet Group. You can choose from which group the snippets should be removed.
- **Load into Cube 2D:** Select at least one snippet to be imported into the Cube 2D View.
- **Load into Canvas:** Select at least one snippet that should be imported into the Canvas View in the Analysis tab (More information see 11.3).

7.3 DATABASE – CATEGORIES

There are eight category groups, which help to categorize the snippet as desired:

- **Snippet Groups:** Snippet groups combine any number of snippets that fit into a desired category. The snippet groups can be loaded into the Canvas View, and all snippets contained therein are displayed accordingly.
- **Material Category:** A material category can be assigned to each snippet. For example, a snippet made of cotton belongs to the material category textiles, pure fibers, or similar.
- **Materials:** Assign a material to each snippet. These are listed here. Each material can also be assigned an ROI color.

- **Owners:** If there are different owners of the snippets, they can be defined. Different editors can also be defined here.
- **Locations:** To assign different locations to the snippets, they can be created here.
- **Application Area:** Define any application areas to which the snippets can be assigned.
- **Background Materials:** Possibility to define different background materials that are visible in the background of the snippets.
- **Verified by:** If it is a verified snippet or is to be assigned to a particular person, any number of names/abbreviations, etc. can be added to this category.

Any number of features can be added to each category, allowing the best possible categorization of the snippets.

A new feature can be **added**, **edited** and **deleted** in each of the seven tabs. You can also select a feature that will be displayed by **default** when a new snippet is created.

Add new Material Edit Material Delete Materials			Default: UNDEFINED	Set default
	Label	Color	Category	
1	WO	96,127,61	Textiles	
2	SI	69,89,127	Textiles	
3	LI	127,127,127	Textiles	
4	Cordura	255,170,127	Textiles	
5	EL	170,255,255	Textiles	
6	PA	255,170,255	Textiles	
7	PES	0,170,0	Textiles	
8	Cotton 100%	255,255,0	Textiles	
9	PVC	170,85,127	Plastics	
10	PET	85,170,255	Plastics	
11	UNDEFINED	No color		

7.4 DATABASE – FILTER

The database **Filter** offers the option of filtering the snippet collection according to the saved categories or capture settings. After automatic application of the filter, only the snippets that match the selected categories are displayed.

Filter:

Snippet Groups (0)

Materials (0)

Background Materials (0)

Owner (0)

Locations (0)

Application Area (0)

Verified by (0)

Detektil

UNDEFINED

Plastics

Textiles

cotton

UNDEFINED

White

Black

UNDEFINED

HAIP Solutions

Customer 1

THN

UNDEFINED

Hannover

Germany

World

UNDEFINED

Research

Optical Sorting

DetekTIL

UNDEFINED

Capture settings

Framerate 0 - 100000

Exposure 0 - 100000

Gain 0 - 100000

Options

Whitebalance

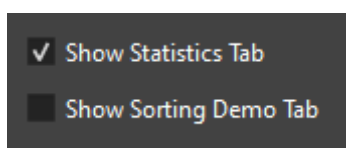
Verified

Reset filter

8 BLACKSTUDIO – STATISTICS TAB

The **Statistics tab** displays statistics for each class, either for live classification or for analysis data, in the form of histograms and tables.

By default, this tab is disabled. To enable it, go to the **Settings tab**, select **Statistics**, and enable **Show Statistics Tab**. After that, BlackStudio must be restarted. Please note that the Statistics tab does not work if **Probability Class Color Mixing**, **Regression** or **PCA Visualization** are enabled in the classification settings.



Switch between the **Live Data** view and the **Analysis Data** view.

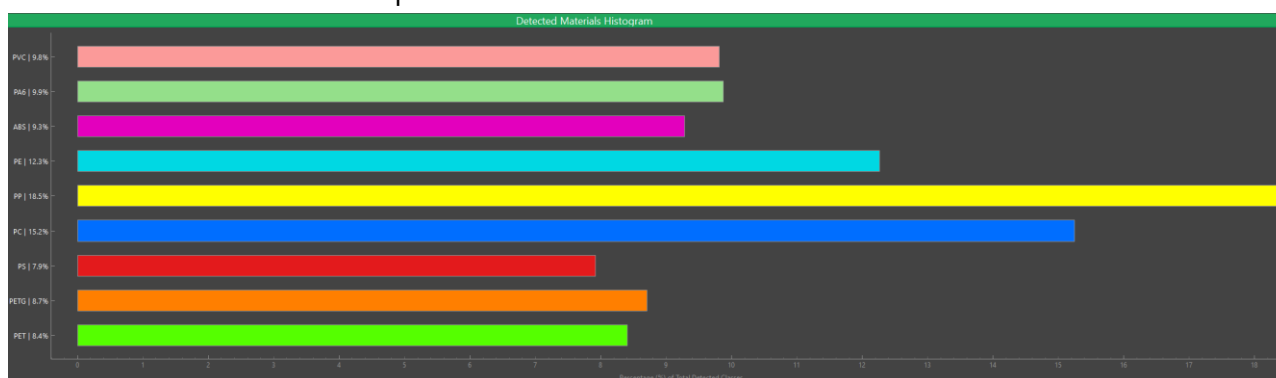
- In the **Live Data** view, the live classification is displayed as a waterfall, along with the corresponding statistical analysis of the detected materials in histogram and tabular formats.
- The **Analysis Data** view displays the statistical analysis of the detected materials from the data cube previously classified in the Analysis tab.



8.1 STATISTICS TAB – DETECTED MATERIALS HISTOGRAM

The **histogram** shows the percentage of pixels for each material class relative to all selected classes. Only the classes of the currently active classifier are considered. Use the **checkboxes** on the left to enable or disable classes for the evaluation.

The histogram can be exported either as a **PNG** or **CSV** file using the dedicated buttons in **Export**. In the **Live Data** view, you can use the **Config** tab to set the time interval (in seconds) over which values are averaged for both the histogram and the table. The shorter the averaging time, the faster the statistics are updated.



8.2 STATISTICS TAB – DETECTED MATERIALS TABLE

The **table** shows, for each selected class, the percentage of pixels relative to the total, as well as the absolute number of pixels per class. The table can be exported as **CSV** using the dedicated button in **Export**. In the **Live Data** view, you can use the **Config** tab to set the time interval (in seconds) over which values are averaged for both the histogram and the table. The shorter the averaging time, the faster the statistics are updated.

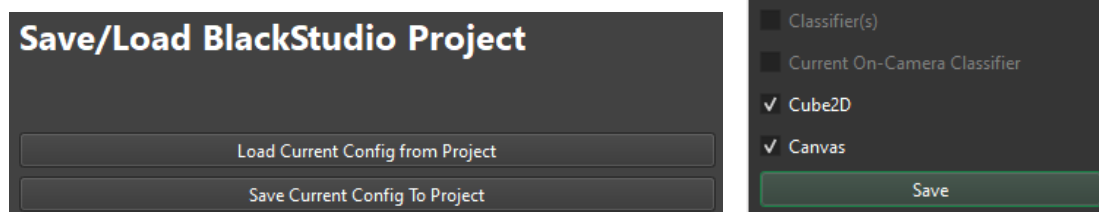
Classes	Config	Export	Class	Percentage	Pixels
☒ PET			1 PET	8.4%	173802
☒ PETG			2 PETG	8.7%	180044
☒ PS			3 PS	7.9%	163764
☒ PC			4 PC	15.2%	315323
☒ PP			5 PP	18.5%	382239
☒ PE			6 PE	12.3%	253524
☒ ABS			7 ABS	9.3%	191895
☒ PA6			8 PA6	9.9%	204188
☒ PVC			9 PVC	9.8%	202923
☐ Background					
☐ Unknown					

9 BLACKSTUDIO – SETTINGS TAB

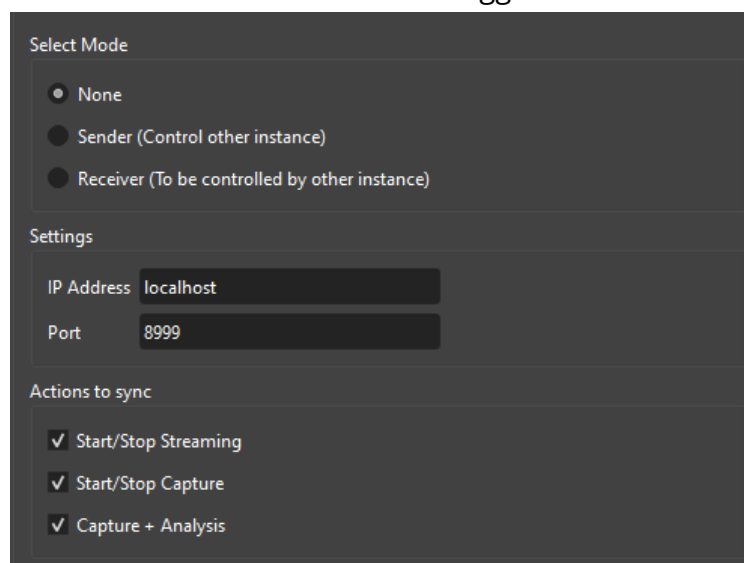
The **Settings tab** in BlackStudio offers the opportunity to make general settings for software use. The settings are divided into the following categories:

- **Project:** This option allows you to save BlackStudio's current state as a config file and reload it to continue working where you left off. Once the file's storage location has been selected, you can use the checkboxes to choose which sections to save:
 - Camera configurations and White-/Blackbalance
 - Classifiers
 - Current On-Camera Classifier
 - Cube 2D View
 - Canvas View

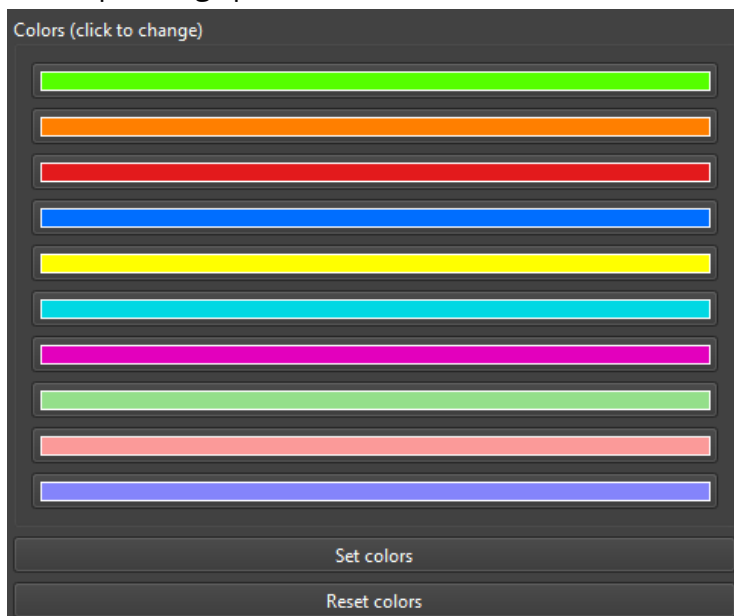
Every time the BlackStudio software is closed, you also have the option to save the project.



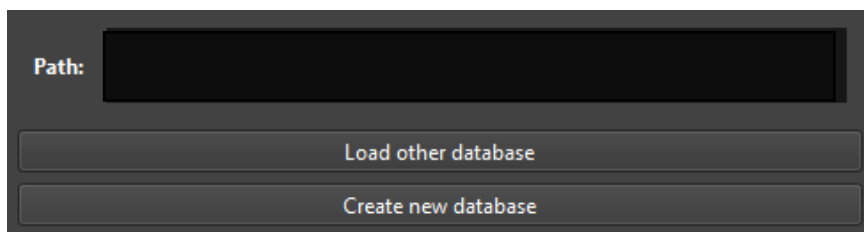
- **About:** Retrieve information about the software version, contact information of HAIP Solutions, etc.
- **Sync:** Settings to synchronize two separate BlackStudio applications so that desired actions will also be activated on the receiver when triggered on the sender.



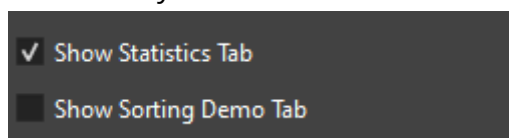
- **Colors:** Settings to change the color coding that is used in BlackStudio for displaying ROIs and the corresponding spectral curves.



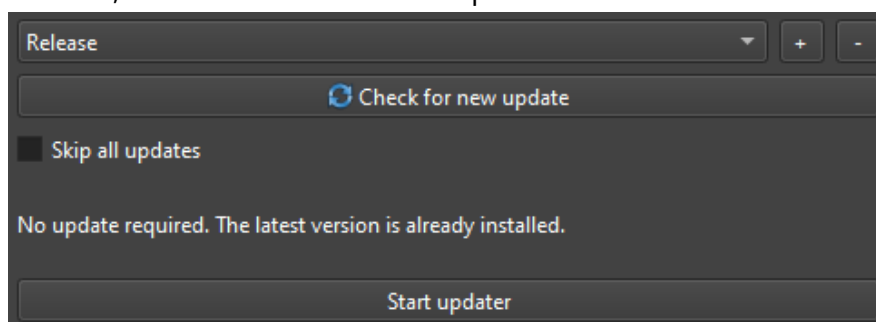
- **Database:** View and change the storage location of the database. Load or create another database.



- **Statistics:** Choose whether the Statistics Tab and/or the Sorting Demo Tab should be displayed. A restart will be necessary.



- **Admin:** Activate Admin functionality.
- **Update:** In this tab, check for new software updates available and install them.



- **Warnings:** Enable or disable “Emit a warning when exposure or frame rate are changed while black balance or white balance are active”.
- **Advanced:** Use these settings to activate advanced functionality. Please note: OpenGL is disabled due to graphical bugs and crashes.

Lazy-loading will only load cube data from file that is actually needed. This will decrease RAM usage and cubes can be viewed faster. Wait times during BlackStudio can be increased in times where data needs to be loaded from disk. Does not always work with Network Drives, so better to load files from local disk.

☐ Activate lazy-loading

OpenGL can improve visualization performance. Disable if it creates artefacts. Note that for changes to take effect, BlackStudio must be restarted.

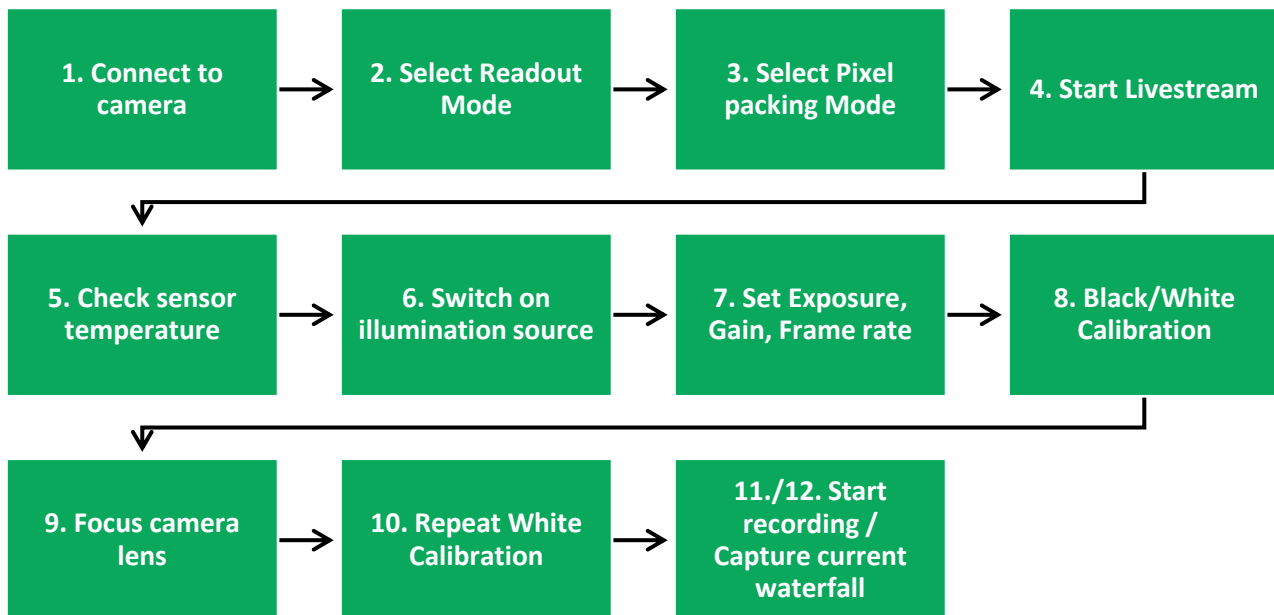
☒ Disable OpenGL

10 DATA ACQUISITION IN BLACKSTUDIO

This chapter describes the most important operations and functionalities that should be considered when acquiring data with BlackStudio. **Please note that not all functionalities in BlackStudio are available for all BlackIndustry camera models.**

10.1 DATA ACQUISITION WORKFLOW

For successful data acquisition with a HAIP BlackIndustry camera and the BlackStudio software, the following basic workflow should be implemented. Further functionalities are explained in later chapters. Data acquisition is performed exclusively in the **Capture tab** of BlackStudio.



1. **Connect to camera:** If the camera is connected via the IP address or another network, the product name, serial number and version number are displayed
2. **Select Readout Mode:** Select a Mode that fits your requirements. The maximum achievable frame rate may change depending on the mode. As soon as the readout mode is changed, new black and white balances must be performed.
3. **Select Pixel Packing Mode:** Select a preferred data size (**8-bit** / **10-bit**). The 10-bit data can be packed for the transmission process (**5/4** or **3/2**) to utilize the Ethernet line efficiently in cost of computational overhead. Alternatively select **Classified Color Mode**, **Regression Float** or **Classified + Spectral** (more information on this in chapter 11.5). If data is lost, this is displayed in the FPS information. The pixel packing mode should be adjusted accordingly.
4. **Start Livestream:** Livestream in all four livestream windows.

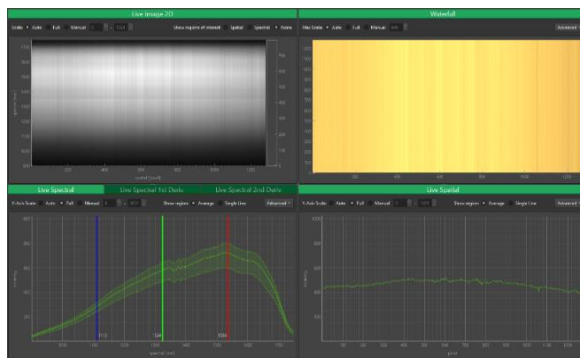
5. **Check sensor temperature:** Spectrometer temperature only for BlackIndustry SWIR 1.7 /Max series.
 - a. **Sensor temperature:** Caution: Changing sensor temperature always affects the spectral data. Always record Black/White Balance after the stabilized temperature is reached.
 - i. The sensor temperature must reach 55°C [131 °F] (temperature display changes from red to green) before recording may begin
 - ii. Once the target temperature has been reached, the camera automatically regulates the temperature level.
 - iii. You can also refer to the blue status LED at the back of the camera to check if the correct sensor temperature has been reached.
 - b. **Spectrometer temperature:**
 - i. Internal spectrometer temperature control. There is no target temperature required.
 - b. **Processor temperature:**
 - i. Internal processor temperature control. There is no target temperature required.
6. **Switch on illumination source** (if not yet done).
7. **Set Exposure, Gain, Frame rate:** Adjust Exposure & Analog Gain settings to achieve an adequate illumination level and to eliminate oversaturation (for 10-bit Pixel Packing, indicated in red). **Caution: The exposure value should always be increased first before the gain value is increased.**
 - a. Test the settings with your samples.
 - b. Live Spectral View: Enable **Show Min/Max** → Ideal **Intensity at max. 80 %**.
 - c. Set the desired **frame rate** in fps/Hz → Maximum frame rate depends on readout mode (max. 1330 Hz). The respective FPS limit is displayed.



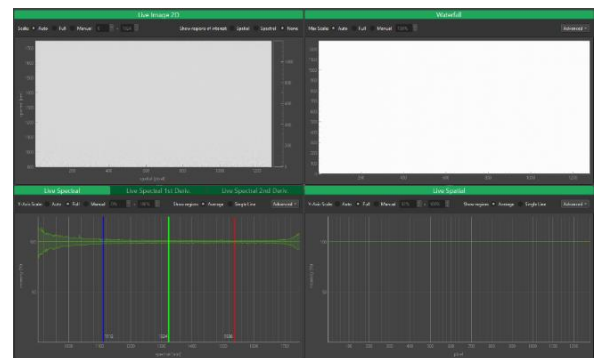
8. Black/White Calibration:

- Black Calibration:** Cover the camera lens entirely with dark object (e.g. lens cap) and press **Record Blackbalance**.
- White Calibration:** Capture an image line from an entirely white surface (e.g. white tile) and press **Record Whitebalance**.
- Each black and white calibration created is automatically saved as a file in the background. If BlackStudio has been closed, the calibrations can be reloaded using **Load last Black/Whitebalance**.

Before Calibration:



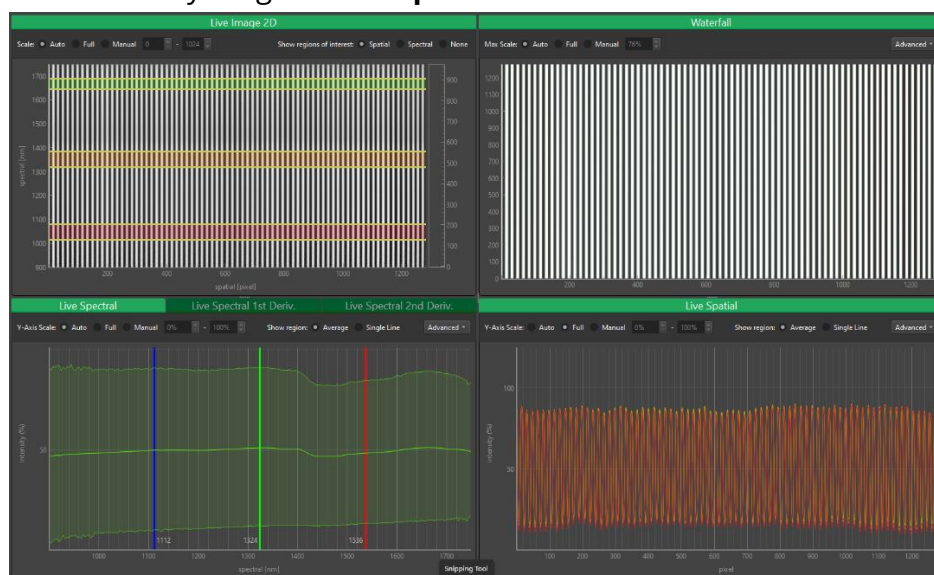
After Calibration:



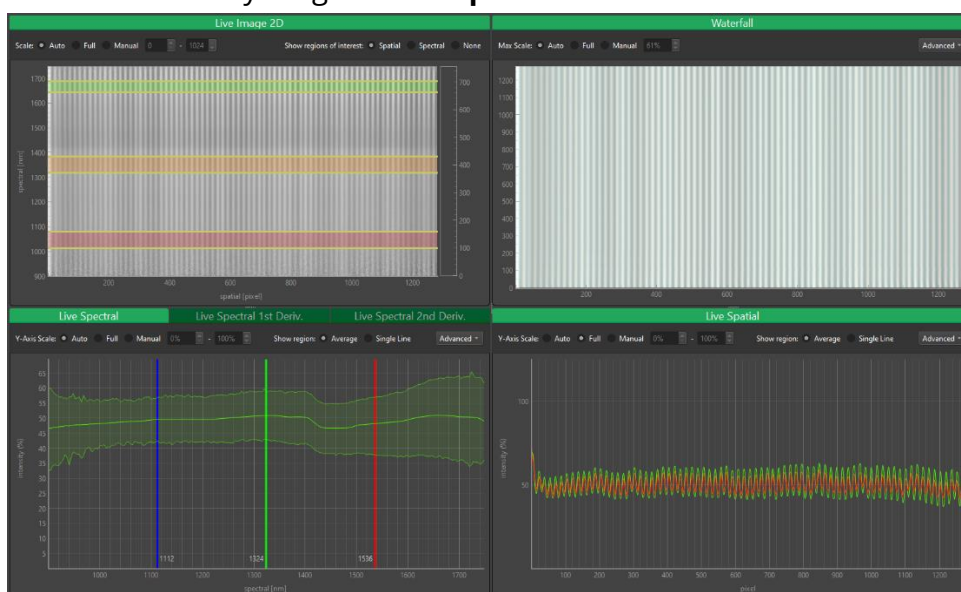
9. Focus camera lens:

- Place a focus control paper sheet under the camera lens so that the distance between camera lens and scanning area remains the same.
- Ensure that the lines are at a ninety-degree angle to the scanning direction of the camera.
- Place 3 spatial ROIs in the **Live Image 2D window** (top, middle, bottom).
- Focus the camera lens manually by turning the wheel on the lens.

Focused: Wide intensity range in Live Spatial View.



Unfocused: Small intensity range in **Live Spatial View**.



10. **Repeat White Calibration:** see step 8.

11. **Start recording:**

- By pressing the button **Start Recording**, each pixel line scanned by the camera is recorded and consecutively chained together.
- This is represented by the **Waterfall window** where each new pixel line is appended from above to the lines already captured.
- Pressing **Stop Recording** stops the recording of the pixel lines, and the data generated is automatically opened in the **Cube 2D View** of BlackStudio.

12. **Capture current waterfall:**

- Pressing the button **Capture current waterfall** captures all currently displayed pixel lines in the waterfall window.
- Afterwards you will be redirected to the **Analysis tab**, where the data is displayed in the Cube 2D window.

10.2 DATA ACQUISITION – PIXEL PACKING

Pixel Packing enables efficient use of the Ethernet bandwidth. Specify whether the pixels are to be sent as 8-bit or 10-bit data and how they should be packed.

Please select the data size and the type of pixel packing via the **Pixel Packing** selection. Alternatively select **Classified Color Mode**, **Regression Float** or **Classified + Spectral** (More information see chapter 11.5).

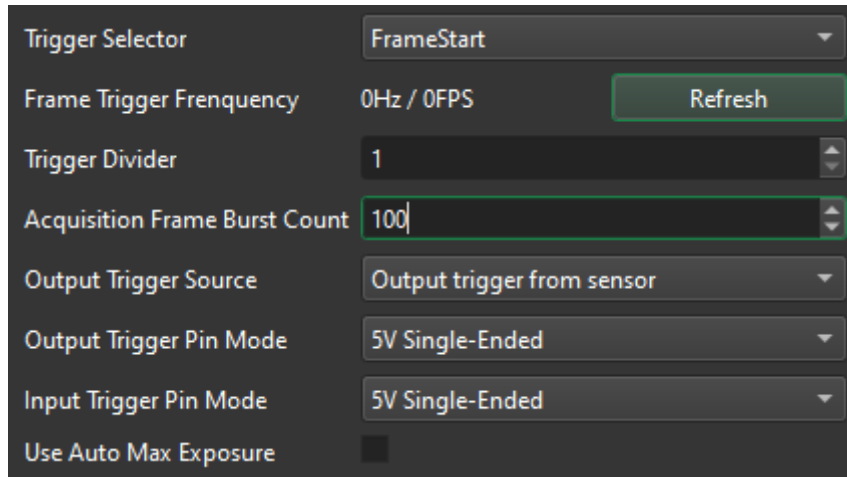
The following options are available:

PIXEL SIZE IN BIT	SENDING DATA IN BYTE	DESCRIPTION	TRADE-OFF
8	1	Reduces the 10-bit values to 8-bit to transfer 1 byte per pixel	Information loss possible
10	2	10-bit values are not reduced and transferred with 2 bytes per pixel	- No information loss - High Ethernet usage
10	5/4	Four 10-bit values are packed into five bytes (40 bits) (5/4)	- No information loss - Reduces Ethernet usage - Higher computational costs on receiver side to unpack the pixels after receiving
10	3/2	Two 10-bit values are packed into three bytes (24 bits) (3/2)	- No information loss - Reduces Ethernet usage - Higher computational costs on receiver side to unpack the pixels after receiving

10.3 DATA ACQUISITION - EXTERNAL TRIGGER FUNCTIONALITY

The HAIP BlackIndustry SWIR 1.7 / Max camera supports external trigger functionality. A trigger is an electrical signal that triggers the sensor and starts the exposure. By using the **Trigger** function, the trigger can be activated from outside the camera. This enables optimal synchronization between the camera and the transmitter of the signal, for example a conveyor belt.

The following **Trigger settings** can be selected in BlackStudio:



The screenshot shows a configuration window with the following settings:

- Trigger Selector:** FrameStart
- Frame Trigger Frequency:** 0Hz / 0FPS (with a Refresh button)
- Trigger Divider:** 1
- Acquisition Frame Burst Count:** 100
- Output Trigger Source:** Output trigger from sensor
- Output Trigger Pin Mode:** 5V Single-Ended
- Input Trigger Pin Mode:** 5V Single-Ended
- Use Auto Max Exposure:** (checkbox is unchecked)

- **Trigger Selector:** Choose between two external trigger modes. Detailed information on both modes in chapters 10.3.1 and 10.3.2.
 - **Frame Start**
 - **Frame Burst Start**
- **Frame Trigger Frequency:**
Displays the currently selected frame rate / frame trigger frequency in Hertz (Hz).
- **Trigger Divider:**
Select the even integer divider for the external trigger signal (1:1 - 1:100).
- **Acquisition Frame Burst Count:**
Select how many frames should be acquired in Frame Burst Start Mode.
- **Output Trigger Source:**
Select the desired signal output. More information on the output options, see chapter 10.3.3.
 - **Output trigger from sensor**
 - **Rising edge output trigger from external**
 - **Falling edge output trigger from external**
- **Output Trigger Pin Mode:**
Please select the desired output pin assignment. More information on Pin assignment see User Manual BlackIndustry SWIR 1.7 / Max camera, chapter 5.5.
 - **5 V Single-Ended**
 - **24 Single-Ended**
 - **Differential**
- **Input Trigger Pin Mode:**
Please select the desired output pin assignment. More information on Pin assignment see User Manual BlackIndustry SWIR 1.7 / Max camera, chapter 5.5.
 - **5 V Single-Ended**
 - **24 Single-Ended**
 - **Differential**

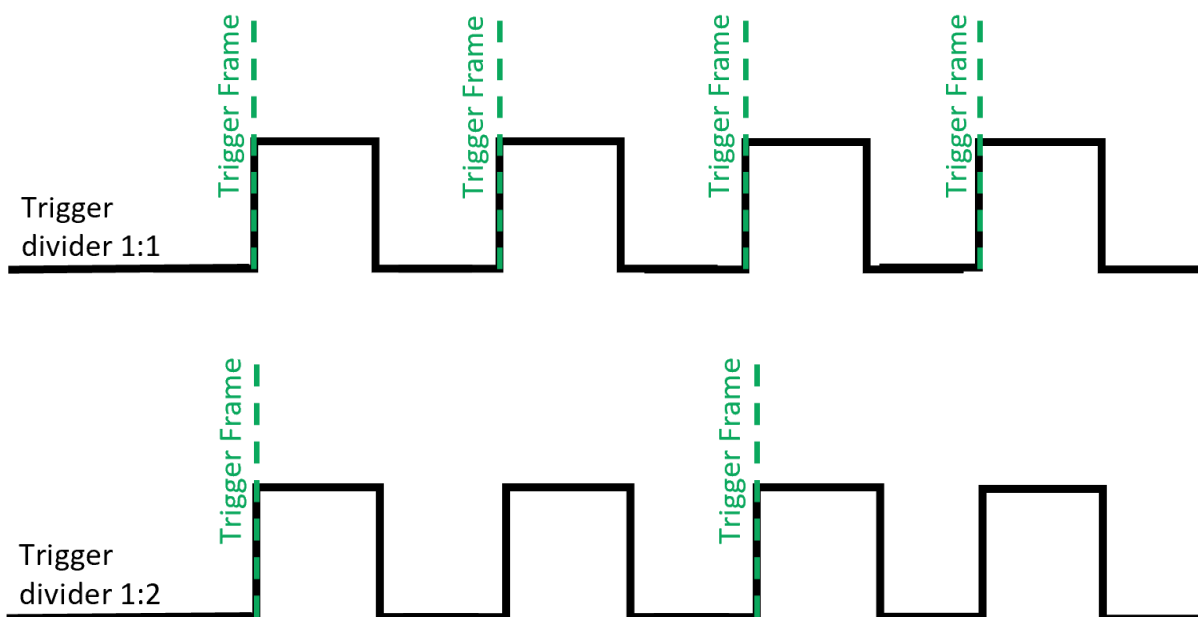
- **Use Auto Max Exposure:**

The maximum value of exposure varies depending on the frame trigger frequency/frame-rate. Activate the checkbox to ensure that the maximum exposure is set automatically. Otherwise, exposure can be set manually.

10.3.1 EXTERNAL TRIGGER - FRAME START MODE

The **Frame Start Mode** enables the integration of a constant external trigger signal into the camera system. Depending on the desired frequency, the sensor is triggered when the edge of the signal rises.

1. The external trigger signal is fed through one of the pins of the M12 connector into the trigger divider of the camera. The divider divides the signal by even integer values or 1 (1, 2, 4, 6, 8, 10, ..., 100). The selectable integer dividers range from 1:1 to 1:100.
2. The processed trigger signal is sent to the sensor to start the exposure.

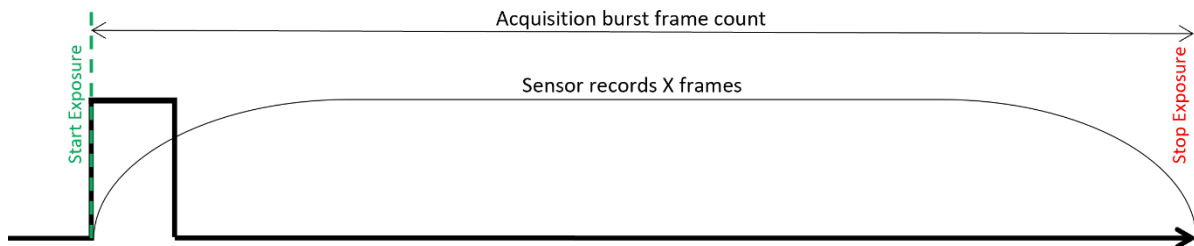


10.3.2 EXTERNAL TRIGGER - FRAME BURST MODE

With the **Frame Burst Start Mode**, an external trigger signal can also be integrated into the BlackIndustry SWIR 1.7 / Max camera. With this mode, it is possible to specify how many frames/lines are to be transmitted. Thus, a series of frames/lines is captured. If, for example, a signal is triggered by an object on a conveyor belt, the trigger of the camera sensor is activated

for a defined interval so that the entire object can be scanned. Once this is done, the transmission stops automatically.

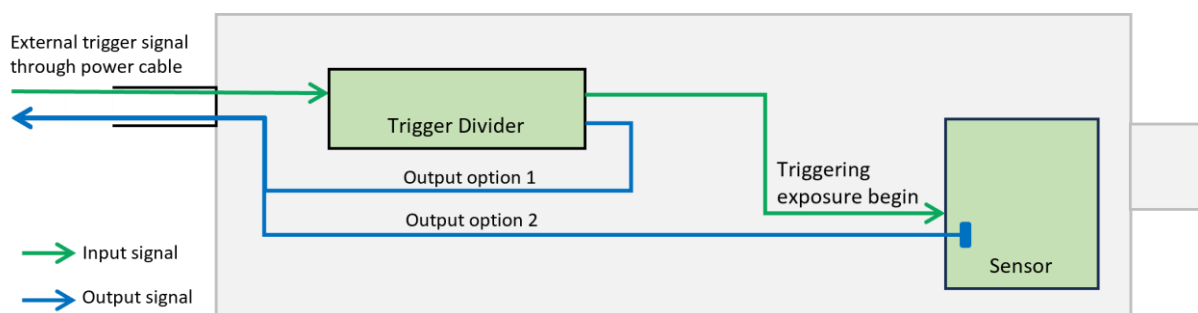
- The external trigger signal is fed through one of the pins of the M12 connector and sent to the camera sensor to directly start the exposure.
 - No trigger divider is utilized in this mode.
 - The frame rate, exposure and gain settings are specified by the user.



10.3.3 EXTERNAL TRIGGER - TRIGGER OUTPUT

In addition to receiving, the BlackIndustry SWIR 1.7 / Max camera can also be used as a transmitter by emitting various trigger signals. You may select your preferred output option:

- **Output trigger from sensor:** The output signal is generated by the camera's sensor. The rising edge signal is output only when the sensor is streaming.
- **Rising edge output trigger from external:** The processed input trigger signal is provided. The rising edge signal is output as soon as an external signal comes in and gets divided.
- **Falling edge output trigger from external:** The processed input trigger signal is provided. The falling edge signal is output as soon as an external signal comes in and gets divided.



10.4 DATA ACQUISITION – MULTI ROI MODES

The BlackIndustry SWIR 1.7 / Max /Pro Max camera has 4 available readout modes. You may select between two **Full modes** and two **MULTI-ROI modes**. By choosing one of the **MULTI-ROI modes** (Mode 2 / Mode 3), you can select multiple **Spectral Regions of Interest** that reduce the number of considered spectral bands and consequently increase the achievable frame rate. The output data is generated either **calibrated** or **uncalibrated**:

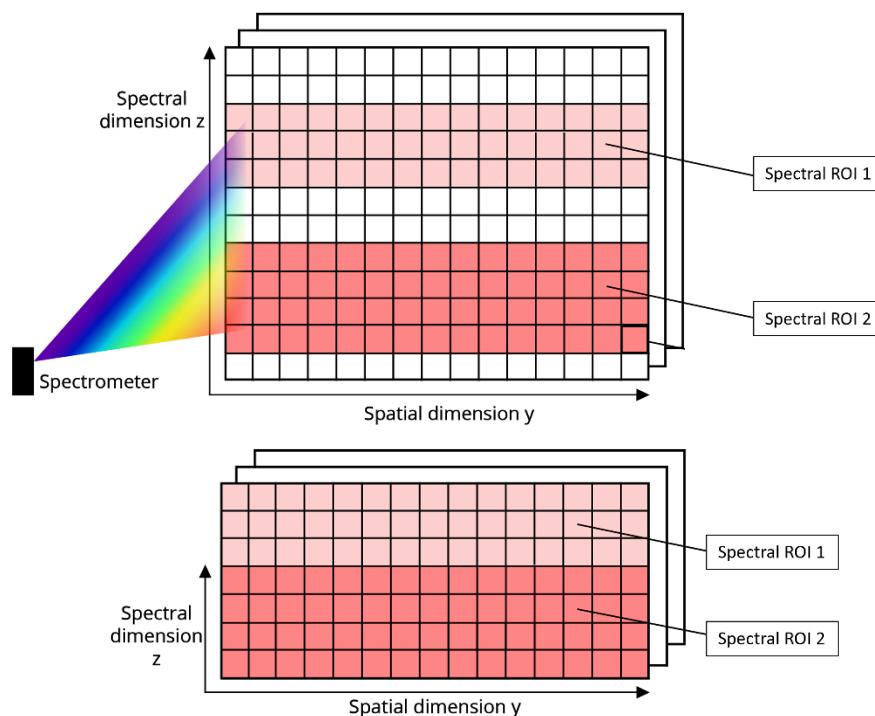
- Mode 2: (Uncalibrated) **MULTI-ROI Mode**
- Mode 3: **CALIBRATED MULTI-ROI Mode**

The other cameras of the BlackIndustry series do not support MULTI-ROI functionality in the strict sense, but they have predefined Spectral Regions modes, each consisting of one specific Spectral Region.

10.4.1 FUNCTIONALITY OF SPECTRAL REGIONS

When using the HAIP BlackIndustry SWIR 1.7 / Max / Pro Max camera, it is possible to increase the frame rate by selecting individual relevant spectral sensor regions. This selection is made by creating **Spectral Regions** that cover the desired sensor regions. The sensor in the camera then only reads the spectral information from the sensor pixels within the Spectral Regions. Sensor pixels outside the Spectral Regions are not captured. For the output, all selected Spectral Regions, and the spectral information they contain are added together.

Illustration of the sensor arrays per spatial pixel line. The red zones display selected Spectral Regions (ROI):



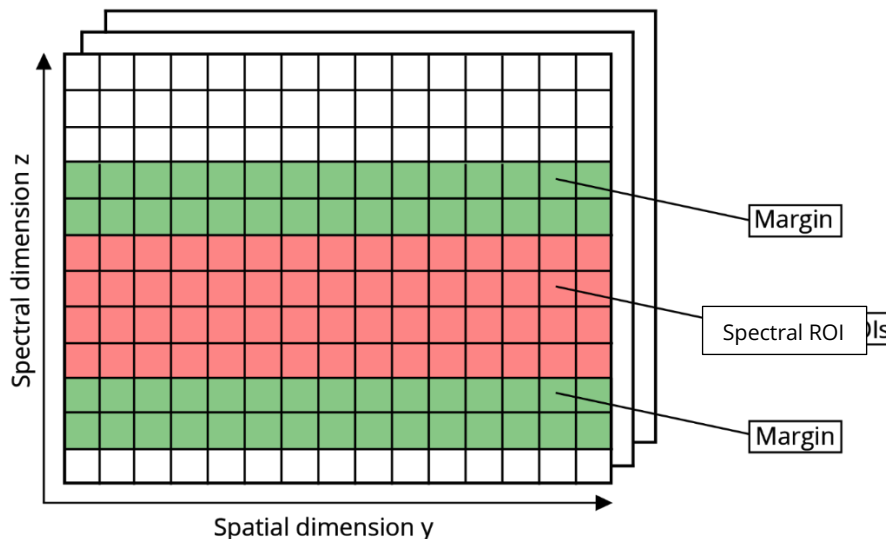
CALIBRATED MULTI-ROI Mode:

If desired, the spectral information is pre-processed internally by the GPU in the camera. The individual sensor pixels are assigned to a specific wavelength band and calibrated using:

- Dead pixel correction
- Smile correction
- Keystone correction

Calibration is essential if the output data is to be compared with data from other cameras!

As the captured spectral information arrives at the sensor in a curved shape, it is necessary that the selected Spectral Region is enlarged by a margin on each side in order to apply the calibration. These margins are used for the internal correction and then deleted, so they are not output. Consequently, the frame rate decreases due to the additional computing power required.



UNCALIBRATED MULTI-ROI Mode:

If a maximum frame rate is required, the spectral information can be output in an uncalibrated state. Consequently, no additional margins are added to the selected Spectral Regions. However, this entails the disadvantage that the output data is not calibrated and therefore cannot be compared with the data from other cameras. It is the customer's responsibility to further calibrate the data, if desired.

10.4.2 SELECT SPECTRAL REGIONS IN BLACKSTUDIO

In order to use the spectral MULTI-ROI functionality of the BlackIndustry SWIR 1.7 / Max cameras, the **Spectral Regions** have to be defined first. This is done via the **Spectral ROI tab** in the **Spectral View** of the **Analysis tab** (see chapter 6.4).

Procedure for uncalibrated Spectral Regions (MULTI-ROI Mode, Mode 2):

Not recommend for use

1. Uncalibrated Spectral Regions can be selected via **Select Manual ROI Regions**.
 - a. Any sensor region of the camera can be selected. If several consecutive regions are selected, one single Region is created from the sensor regions.
 - b. Please note that the regions differ depending on each specific camera (therefore the Regions are uncalibrated).
 - c. A maximum of 8 Regions can be created in total. The number of sensor regions selected does not matter.
2. To confirm the selected regions and transfer them to the camera, press **Set ROIs X/8**.
3. Alternatively, use **Auto select ROIs from calibrated ROI settings** to copy ROIs that were previously set on the camera to the manual selection. BlackStudio attempts to automatically find the matching regions of your camera (please note that ROIs must first be selected and set via the **Calibrated ROIs** section).
4. Click the **Load wavelength table from camera** button to open a detailed table showing which wavelengths strike which sensor areas. For more information and the corresponding technical note, contact HAIP Solutions. This feature is available from BlackIndustry SWIR 1.7 firmware version 1.1.9.
5. Set **mode 2 MULTI-ROI** in the **camera mode settings** in the **Capture tab**.
 - a. The maximum achievable frame rate is then calculated automatically (see **Camera settings** in the **Capture tab**).
6. Perform a Black- & White Calibration.
7. Start data acquisition.

Procedure for calibrated spectral ROIs (CALIBRATED MULTI-ROI, Mode 3):

1. Calibrated Spectral Regions can be set and visualized by activating **Activate classifier regions** or **Get Calibrated Regions from Camera**.
 - a. By clicking in the **Spectral View** window, Spectral Regions and Classifier Regions can be created, displayed as red bars. The size of the Regions can be changed by dragging the sliders.
 - b. A maximum of 8 Regions can be created in total. The number of selected sensor regions within the Spectral Regions is displayed at the bottom left at **Regions used**.

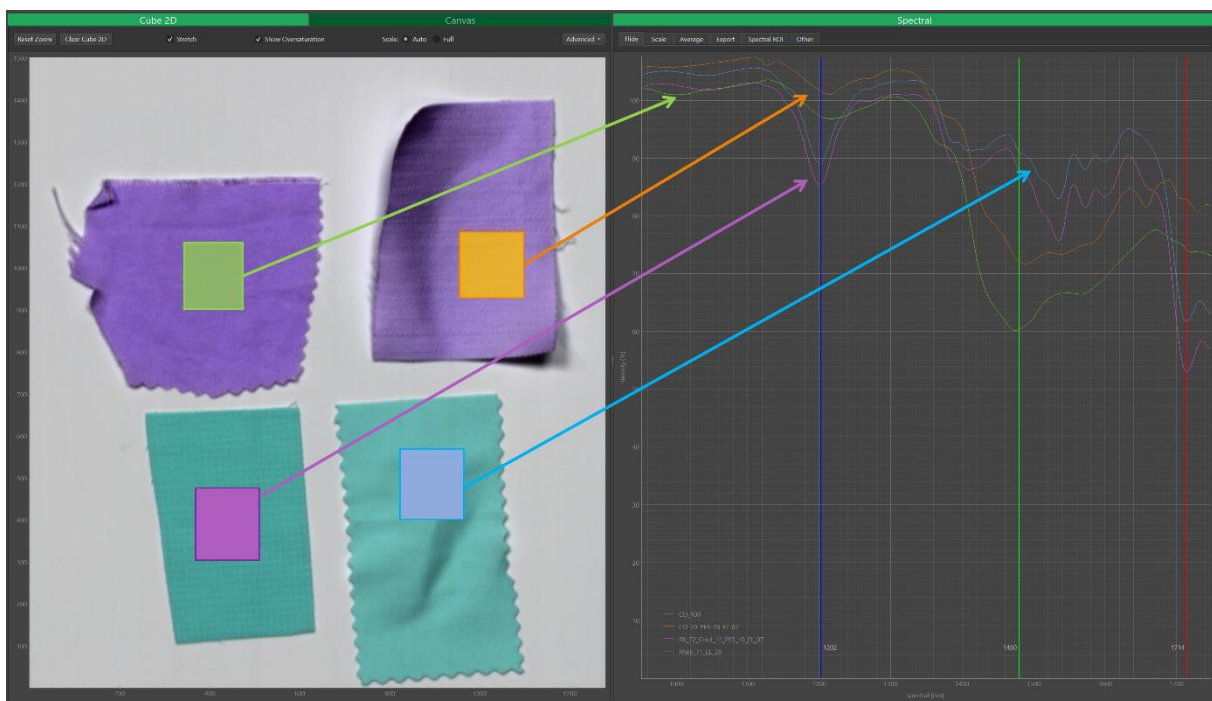
2. Once all desired Regions are created, press **Write Calibrated Regions to Camera** to transfer the Spectral Regions to the camera. The Regions are stored on the camera until new ones are selected and set. Even rebooting the camera does not delete the Spectral Regions.
3. If you want to access the Spectral Regions stored on the camera, press the **Get Calibrated Regions from Camera** button.
4. Finally, select **Mode 3 CALIBRATED MULTI-ROI** in the **camera settings** in the **Capture tab**. The achievable frame rate is then calculated automatically.
5. Perform a Black- & White Calibration.
6. Start data acquisition.

11 DATA ANALYSIS IN BLACKSTUDIO

This chapter describes the most important operations and functionalities that should be considered when analyzing data with BlackStudio. Please note that not all functionalities in BlackStudio are available for all BlackIndustry camera models.

11.1 DATA ANALYSIS – SETTING SPATIAL ROI MARKERS

Creating a **spatial region of interest (ROI) marker** in the **Cube 2D View** or in the **Canvas View** in the **Analysis tab**, enables visualization and analysis of the spectral information from the selected area. The spectral information is displayed in the **Spectral View** as intensity graph, in the same color as the ROI marker. Based on the created and saved ROI markers, classifications can be implemented, and further analyses can be performed.



The following actions are available when managing ROI markers:

- **Double click:**
 - In **Cube 2D View** or **Canvas View**: Creates a new ROI marker with a new color. A new intensity curve appears in the **Spectral View**.
 - On an existing ROI marker: Duplicates the ROI marker. Nevertheless, only one spectral intensity curve is displayed in the **Spectral View**.
- **Right click:** Right-click on an existing ROI marker to delete it.
- **Turning wheel:** Click on the turning wheel on the left border of a ROI marker and move the mouse to rotate the ROI marker.

- **Corners:** Click on a corner of a ROI marker to change its size.
- **Drag and drop:** Move the ROI marker by drag and drop.
- **Show ROIs:** To hide all ROI-classes, uncheck **Show ROIs** in the **Analysis Settings**.
- **Save capture +ROIs:** Press the button in the analysis settings to save the hyperspectral data cube including the created ROI markers. Otherwise, the ROI markers will be lost.

11.2 DATA ANALYSIS – PREPARE DATABASE AND CREATE A SNIPPET

Before data snippets are added to the **database**, the following steps should be considered first:

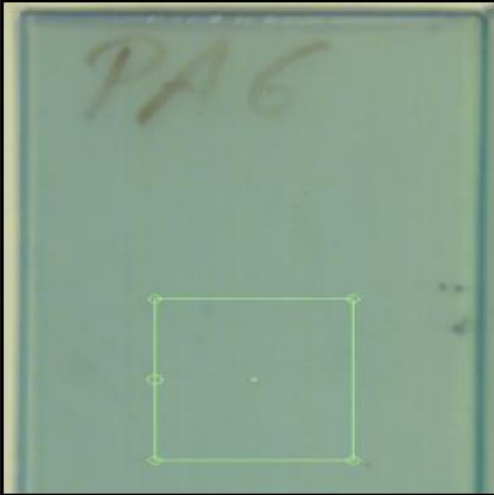
1. Select a suitable **storage location** for the **database**. This location can also be on a network drive or USB flash drive. The current storage path can be checked in the **Database settings** (Settings Tab → Database Settings).
2. **Add content** to the seven database categories that are relevant to you (e.g. Material Category → Add new material category → 'Plastics').
 - a. It is advisable to start with the category on the far right when creating the groups and to proceed further to the left.

The database is now prepared.

3. Capture data via the **Capture Tab** or import previously acquired data in the **Cube 2D View**.
4. If desired, create **ROI markers** of relevant data within the **Cube 2D View**.
5. Press the button **Export to database**, a snippet window will open:
 - a. Via **Select Area**, a snippet can be created by manually defining the spatial extent. Alternatively, the entire image can be saved as a snippet via **Select Full Image**.
 - i. It is not mandatory to include a ROI marker in the snippet, but it is advantageous, as the spectral information within the ROI will also be stored (It is also possible to add an ROI at a later time).
 - b. Define a snippet **name** and add a **description**, if desired.
 - c. Select a previously created **content category for six database categories** (Material, Background material, Owner, Location, Application Area, Verified by).
 - i. The **+** on the right-hand side can be used to add content categories that do not yet exist in the database categories.
 - d. If the snippet is an object that has already been identified and verified, the **Verified** checkbox can be activated.
 - e. Press **Add** to add the snippet to the database.
 - f. The snippet will be displayed in the **Snippet Collection** Tab in the **Database**.

Export to database

Preview



Select Area

Select Full Image

Name

PA6

Description

PA6 Reference Sample

Material

PA

+

Background Material

White

+

Owner

HAIP Solutions

+

Location

Hannover

+

Application Area

Research

+

Verified

☐

Verified by

UNDEFINED

+

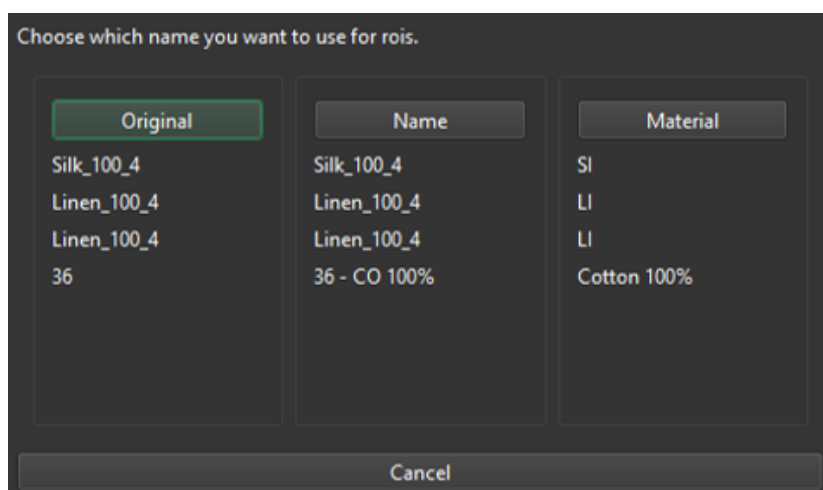
Add

11.3 DATA ANALYSIS – LOAD SNIPPETS TO CANVAS

The **Canvas View** was developed to expand the capabilities of the database. Any number of snippets can be loaded into it and arranged as desired. The spectral curves of loaded or newly created ROI markers are displayed in the spectral window. The ROI markers can also be used to train a classifier (see chapter 11.4).

Please proceed as follows:

1. Open the database.
2. Select the desired snippets by clicking on the respective rows (indicated in green) or select a Snippet Group.
3. Press the button **Load into Canvas**.
 - a. Choose which name you want to use for the ROIs by clicking either on **Original**, **Name** or **Material**.



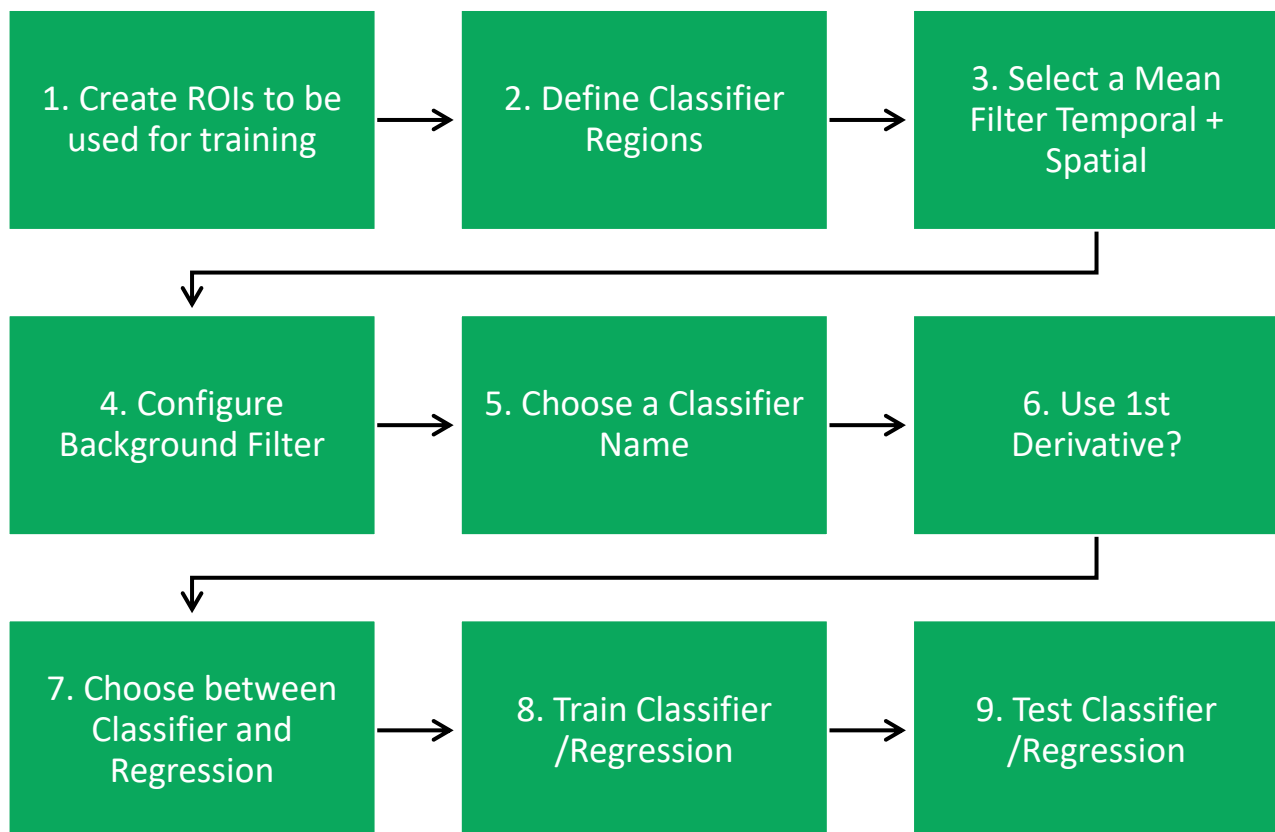
4. You will be re-directed to the **Canvas View** in the Analysis tab.
 - a. If there are still unwanted spectral curves in the spectral window, these can be deleted using the **Clear Cube 2D** button in the **Cube 2D View**.
5. The snippets can be freely arranged and sorted in the canvas window without spatial reference to each other.
 - a. The images and the ROI markers can be moved separately from each other.
6. New ROI markers can be created at any time. If the changes are to be stored in the database, press **Save Snippets ROIs to DB**.
7. The ROI markers can be used as training data for the classifier.
8. The **Clear Canvas** button removes all the contents of the window. However, the snippets are still saved in the database.
9. Save a Canvas View to share it with others via **Save Canvas**.

11.4 DATA ANALYSIS – TRAIN A CLASSIFIER

The **Analysis tab** of BlackStudio offers the possibility to train a specially developed classification algorithm based on self-created ROI markers. These can be created on previously acquired data in the **Cube 2D View** or from snippets in the **Canvas View**. In addition, either the entire wavelength range can be considered for classification or just specifically defined Spectral Regions, so-called Classifier Regions. Which functions are available and how to proceed is explained in this chapter.

11.4.1 GENERAL CLASSIFICATION WORKFLOW

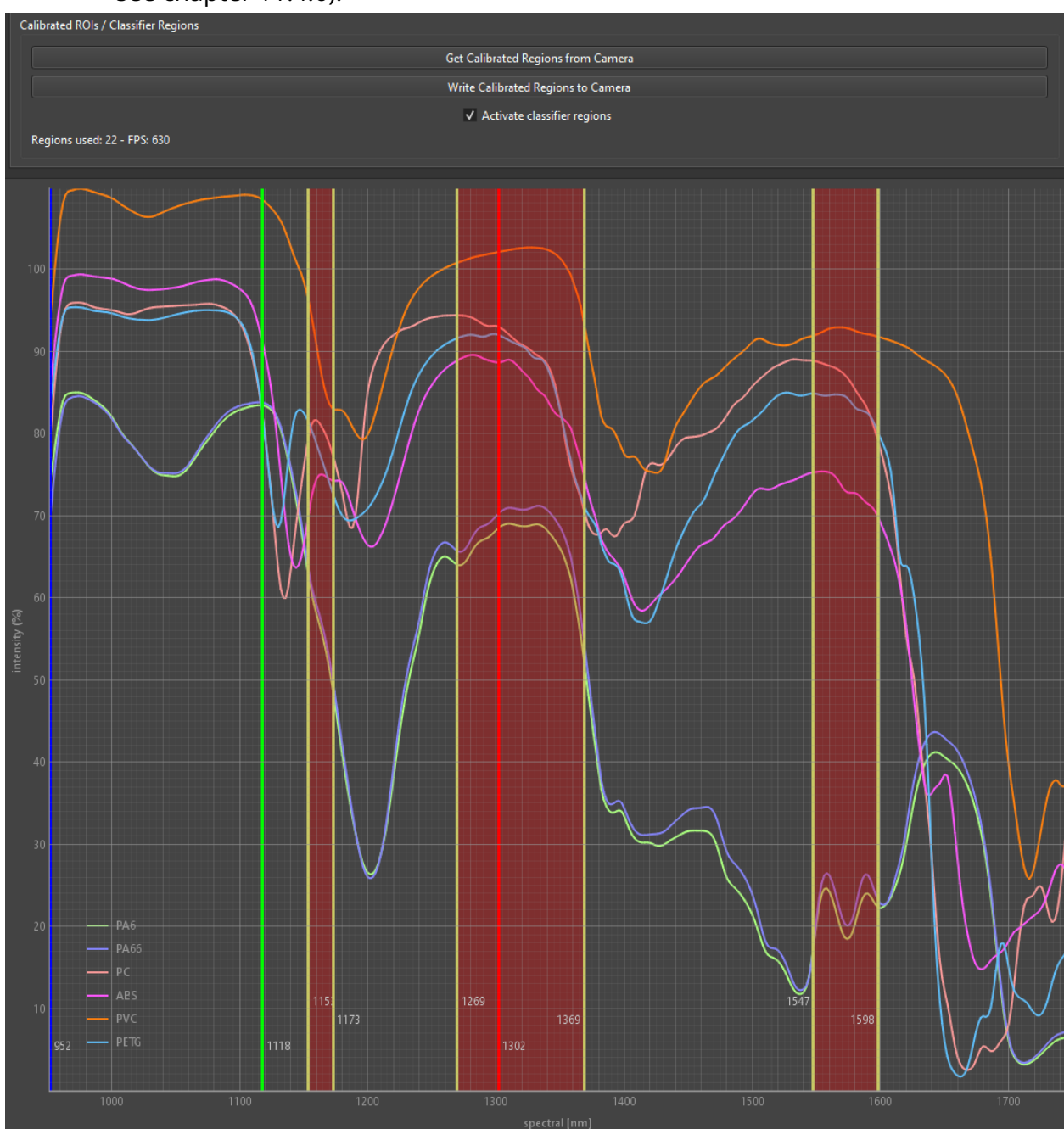
Follow the workflow presented here to implement a successful classification.



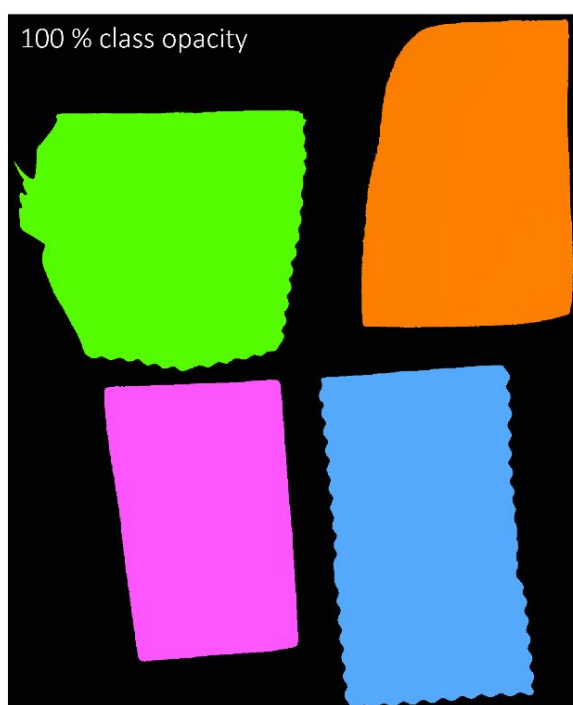
1. **Create ROIs to be used for training:** The basis of a classification are spatial ROI markers, which can be created in the **Cube 2D View** or in the **Canvas View** (see chapter 8.1). The classes that are created in this way are included in the subsequent classification. The more precise the definition of the ROI marker along with the spectral information it contains, the better the classification result will be in the end. There is no maximum or minimum number of ROI markers required for classification.
2. **Define Classifier Regions:** In the **Spectral ROI** tab of the Spectral View, specific Spectral Regions from the wavelength range of the camera can be selected. Only the spectral

information from these regions is considered for the creation of a classifier as well as for creating a background filter.

- Classifier Regions can be activated by enabling **Activate Classifier Regions** and created by double-clicking in the **Spectral View**. Their size can be adjusted as required by dragging the borders. Right-click on a Classifier Region to delete it. All Classifier Regions created are displayed in red.
- To transfer the selected classifier regions to the camera, click **Write Calibrated Regions to Camera**.
- To help assess which wavelengths might be particularly relevant for your classification, you can use the **Visualize Wavelength Importance** tool (for more information, see chapter 11.4.6).



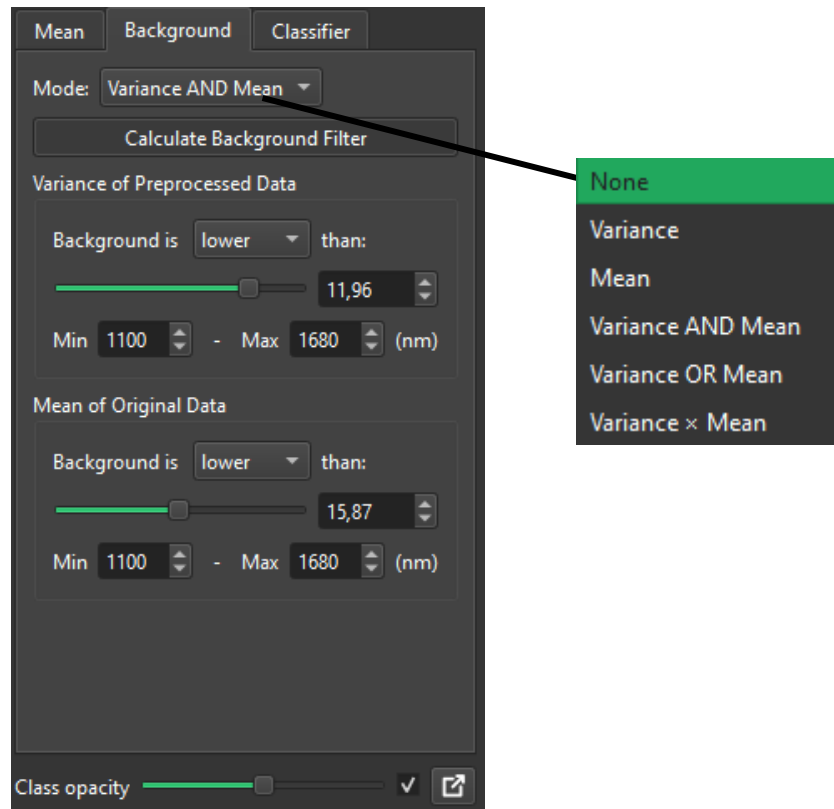
3. **Select a Mean Filter Temporal + Spatial:** By setting a mean filter in the temporal or spatial direction, pixel values in a defined neighborhood (3, 5, 7 or 9 pixels) are replaced by their mean values. This leads to an effective noise reduction.
4. **Configure Background Filter:** A background filter can be created so that only those areas of the **Cube 2D View** or **Canvas View** that have not been marked as background are taken into account in the classification. For more information, see chapter 11.4.2.
5. **Choose a Classifier name:** If the classifier shall be saved externally afterwards, a meaningful name is recommended.
6. **Use 1st Derivation:** Activate the checkbox to pre-process data with the 1st derivative before the classification (Recommended to activate).
7. **Choose between Classifier and Regression:**
 - a. **Classifier:** You may train a classifier using these algorithms: LDA, SVM, SVM+, Logistic, PLS-DA. More information see chapter 11.4.3.
 - b. **Regression:** Currently, support vector regression (SVR) can be selected. For more information, see chapter 11.4.4
8. **Train Classifier/Regression:** Press to train the classification algorithm or regression on the basis of the configured settings. Depending on the data size, this may take a few seconds. Training is completed as soon as the **Save classifier** and **Test classifier** buttons below are no longer grayed out.
9. **Test Classifier/Regression:** Press to apply the classification algorithm or regression to the entire data set in the **Cube 2D View** or to the snippets in the **Canvas View**. The **Class opacity** slider can be used to select the degree of transparency of the classification result. The checkbox on the right-hand side can be used to switch the classification layer on and off.



11.4.2 BACKGROUND FILTER

The background filter creates a background class in which all pixels whose values fall below a defined threshold value are categorized. The pixels within this background class are not taken into account for the actual classification. The background filter is a threshold-based background classifier that utilizes either the mean value of the original spectrum or the variance of the pre-processed spectrum. It is also possible to combine both values.

Attention: If classifier regions were previously created, only data from these areas will be processed in the background filter.

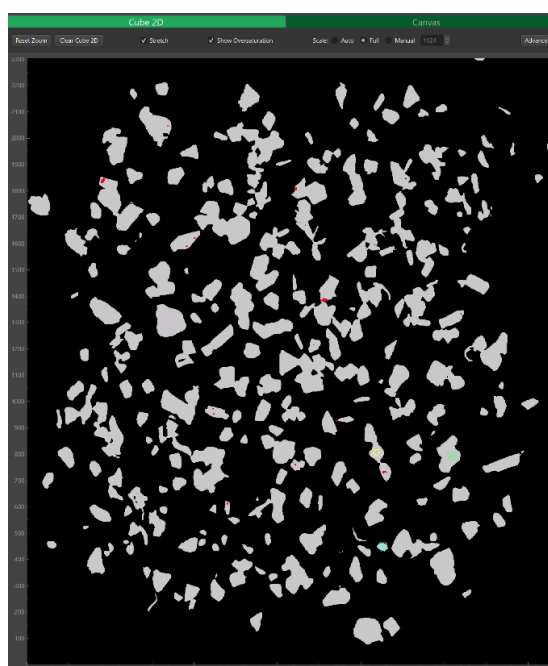
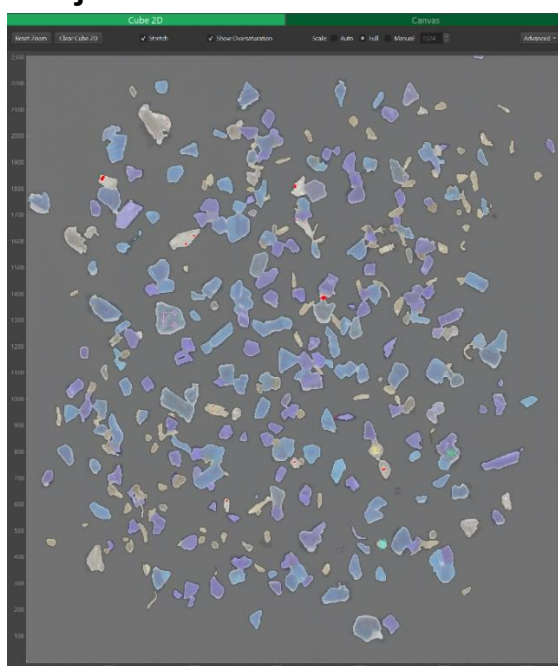


Configure Background Filter:

1. Select a **Mode**:
 - a. **None**: No mode.
 - b. **Variance**: If the variance of pre-processed spectrum is below the threshold value, the pixel is assigned to the Background class.
This mode is to be used with a bright/white background!
 - c. **Mean**: If the mean of original spectrum is below the threshold value, the pixel is assigned to Background class.
 - d. **Variance AND Mean**: If both, variance value and mean value are below the threshold value, the pixel is assigned to Background class.
 - e. **Variance OR Mean**: If either variance value or mean value is below the threshold value, the pixel is assigned to Background class.
 - f. **Variance x Mean**: If the product of the variance value and the mean value is below a threshold value, the pixel is assigned to the background class.

2. Depending on the Mode, change the settings of **Variance of Preprocessed Data** and/or **Mean of Original Data**.
 - a. The corresponding threshold value can be adjusted using the slider or the number display.
 - b. For both values, a wavelength range can be determined in which the values are included. Outside of this range, the values are not taken into account. The default wavelength range is 1100 - 1680 nm.
 - c. **Attention:** Make sure that the set wavelength range matches the classifier regions that might have been created. The more available data the filter has, the more accurate the result will usually be.
3. Press **Calculate Background Filter** to calculate the background filter with the currently set threshold values for variance and mean.
 - a. Depending on the image size, it may take a few seconds.
 - b. The background filter is displayed in the **Cube 2D View** or **Canvas View**. The darker areas are marked as background. The light areas should mark the objects of interest (see figure below).
 - c. The opacity of the background filter display can be changed using the **Class opacity** slider.
4. The background filter can now be modified in real time when changing the threshold values for variance and mean (see step 2).
5. When the background filter is ideally set, the window can be closed.
6. Activate the **Use Background Filter for Classification** checkbox and continue with the creation of the classifier.

Screenshot of the Cube 2D view on which a background filter is being adjusted so that the objects to be classified later are not marked:



11.4.3 CLASSIFICATION ALGORITHMS

The following classification algorithms are available:

- **Linear Discriminant Analysis (LDA):** Linear Discriminant Analysis is a classification method that finds linear combinations of features that best separate different classes. It works by maximizing the distance between class means while minimizing the variance within each class.
 - Wikipedia: https://en.wikipedia.org/wiki/Linear_discriminant_analysis
 - Scikit-Learn: https://scikit-learn.org/stable/modules/lda_qda.html
- **Support Vector Machine (SVM):** A Support Vector Machine finds a decision boundary (hyperplane) that separates classes while maximizing the margin between the closest data points of each class. In BlackStudio a linear SVM is used.
 - Wikipedia: https://en.wikipedia.org/wiki/Support_vector_machine
 - Scikit-Learn: <https://scikit-learn.org/stable/modules/svm.html>
- **Support Vector Machine + Visualize Unknown Class (SVM+):** SVM+ is the same as SVM but trains an extra One-SVM for each class to detect if pixels really belong to that class. It learns the boundary of normal data and identifies new data points that fall outside this boundary as anomalies.
 - Wikipedia: https://en.wikipedia.org/wiki/One-class_classification
 - Scikit-Learn: https://scikit-learn.org/stable/modules/outlier_detection.html#one-class-svm
- **Logistic:** Logistic Regression is a statistical model, despite its name, commonly used for classification. It predicts the probability that a data point belongs to a specific class using a logistic (sigmoid) function.
 - Wikipedia: https://en.wikipedia.org/wiki/Logistic_regression
 - Scikit-Learn: https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
- **Partial Least Squares-Discriminant Analysis (PLS-DA):** PLS-DA is a classification method based on Partial Least Squares regression. It reduces the data to a small number of latent variables that capture both the variance in the predictors and the separation between classes.
 - Wikipedia: https://en.wikipedia.org/wiki/Partial_least_squares_regression
 - Scikit-Learn: https://scikit-learn.org/stable/modules/cross_decomposition.html#partial-least-squares

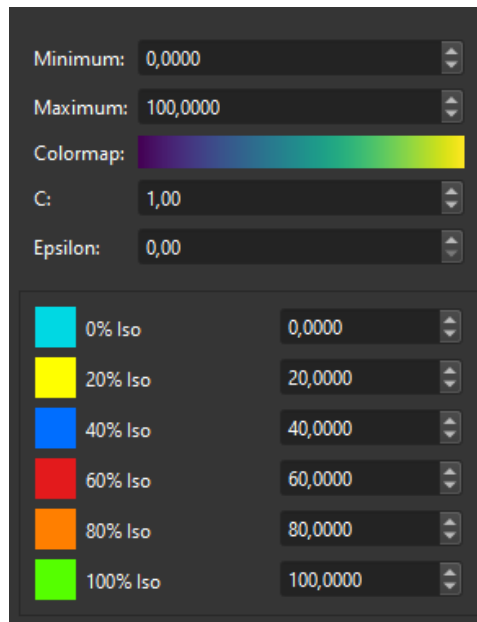
11.4.4 REGRESSION

In addition to various classification algorithms, support vector regression (SVR) can also be performed. Support Vector Regression is a regression algorithm based on Support Vector Machines. It predicts continuous values by finding a function that fits the data while allowing a small margin of error. SVR aims to keep the model as simple as possible while still accurately describing the relationship between the input features and the target value.

- Wikipedia: https://en.wikipedia.org/wiki/Support_vector_machine#Regression
- Scikit-Learn: <https://scikit-learn.org/stable/modules/svm.html#svm-regression>

The following workflow must be considered for a successful regression:

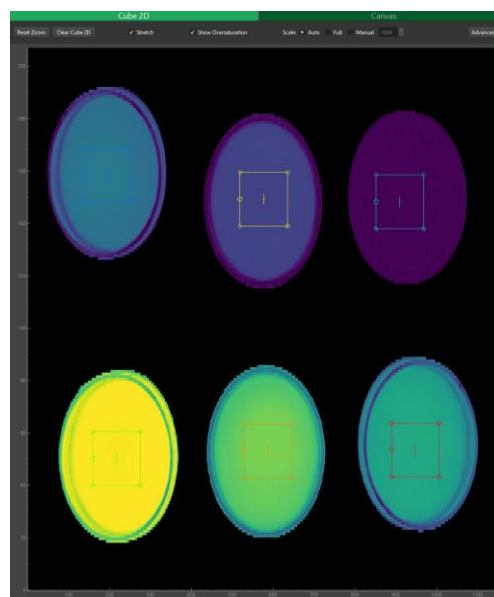
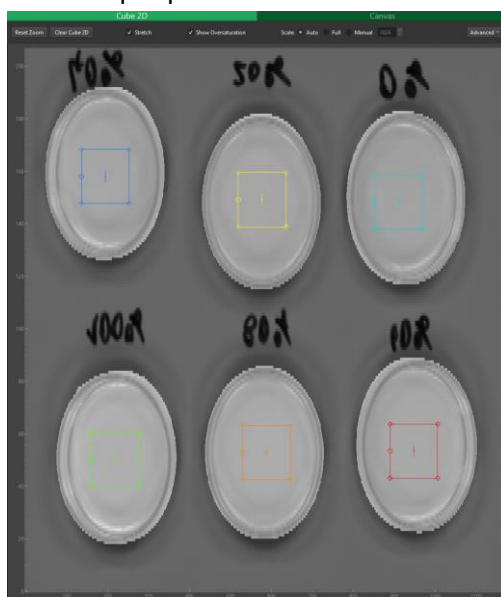
1. Capture data or load a data cube into the Analysis tab. Set spatial ROIs based on which the regression is to be trained.
2. Select **Regression** in Classifier Analysis Tab.
3. Press **Set Regression Values**:
 - a. Select the **minimum** and **maximum** values as well as **C** and **Epsilon** for the regression.
 - b. Assign values corresponding to the ROIs that need to be considered.



Minimum:	0,0000
Maximum:	100,0000
Colormap:	
C:	1,00
Epsilon:	0,00
0% Iso	0,0000
20% Iso	20,0000
40% Iso	40,0000
60% Iso	60,0000
80% Iso	80,0000
100% Iso	100,0000

4. Press **Train classifier**.
5. Press **Test classifier**.

Cube 2D in grayscale with Petri dishes containing liquids with different concentrations of isopropanol: Cube 2D with regression result as color coding:

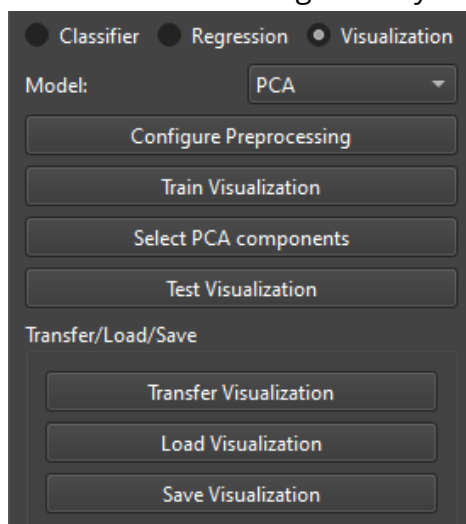


11.4.5 VISUALIZATION

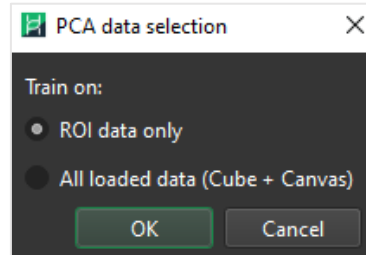
Principal Component Analysis (PCA) can be used to transform hyperspectral image data into an RGB image. This is useful for visualizing and exploring hyperspectral data, or for using the generated RGB image in subsequent image-processing workflows.

The PCA algorithm calculates the principal components that best represent the spectral variation in the data. The software computes the first **8 principal components**. To create an RGB image, you can assign one principal component to each of the red, green, and blue color channels.

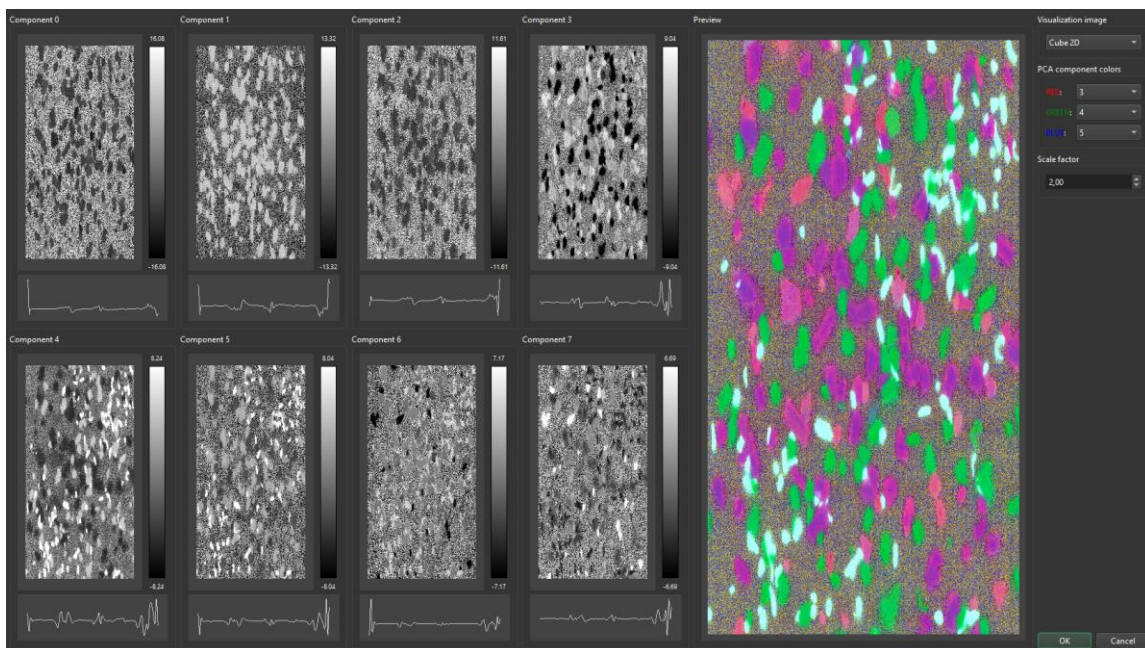
PCA can be applied with or without defined ROIs. For training the PCA transformation, you can choose whether to use the data from the entire image or only the data inside the selected ROIs.



1. Select **Visualization**.
2. If necessary, configure the data **preprocessing** according to your preferences.
3. Press **Train Visualization** to train the PCA model. Choose whether to use the data from the entire image or only the data inside the selected ROIs for training.



4. After completion, press **Select PCA components** to open the dedicated window.
 - a. The first 8 components of the selected image are displayed.
 - b. Visualize any three components using the **red**, **green**, and **blue** channels. To do this, select the component number corresponding to the desired channel on the right-hand side.
 - c. If desired, change the **Scale factor** to manually adjust color intensity of the resulting image.
 - d. A preview of the output is displayed on the right. Once it meets the requirements, click **OK**.

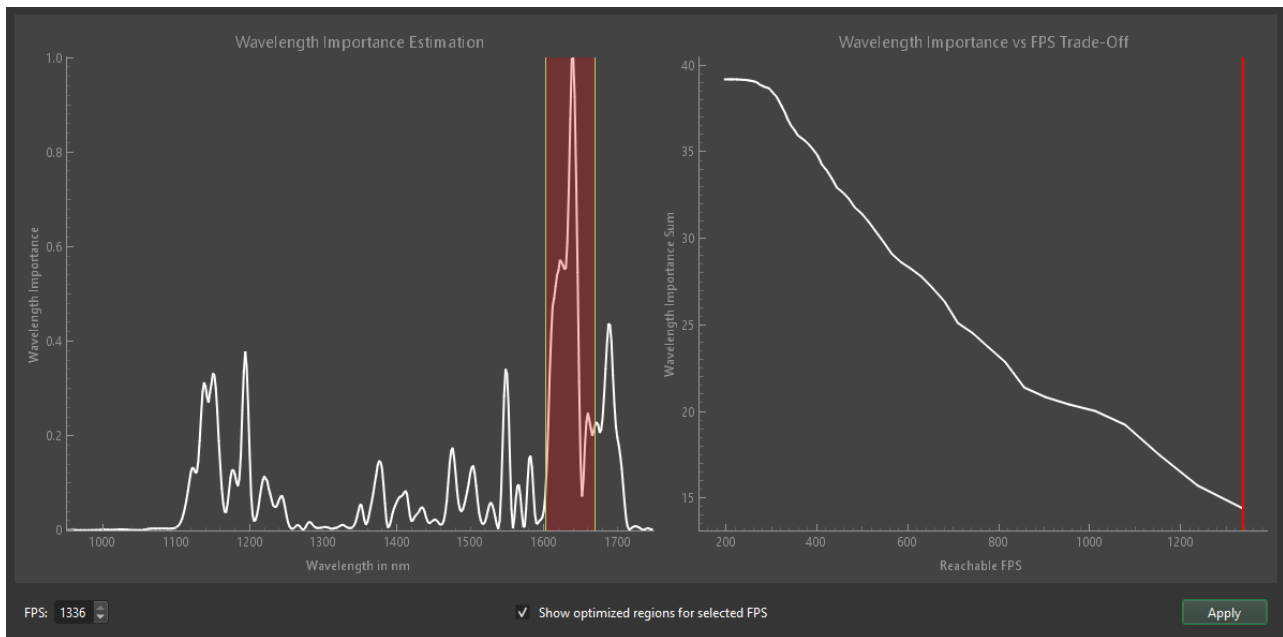


5. Press **Test Visualization** to apply the PCA visualization with the three selected components to the entire **Cube 2D** View and **Canvas** View.
6. Adjust the **opacity** of the visualization using the slider.
7. If desired, **save** and **load** this trained PCA visualization.
8. The trained and configured PCA can then be transferred to camera using **Transfer Visualization** and can be received using the **Classified Color** mode.

11.4.6 VISUALIZE WAVELENGTH IMPORTANCE

Visualize Wavelength Importance is a tool that provides an evaluation of which wavelength ranges may be particularly relevant for the classification of the set ROIs. It therefore serves as a guide when selecting classifier regions.

Once the calculation is complete, a new window opens with two graphs. The left-hand graph shows the importance per wavelength, including suggested classifier regions. The graph on the right-hand side plots the total wavelength importance against the achievable FPS. The desired FPS can be changed using the FPS display at the bottom left of the window. The red marker in the right-hand graph moves accordingly. The less the reachable FPS, the more classifier regions are suggested.

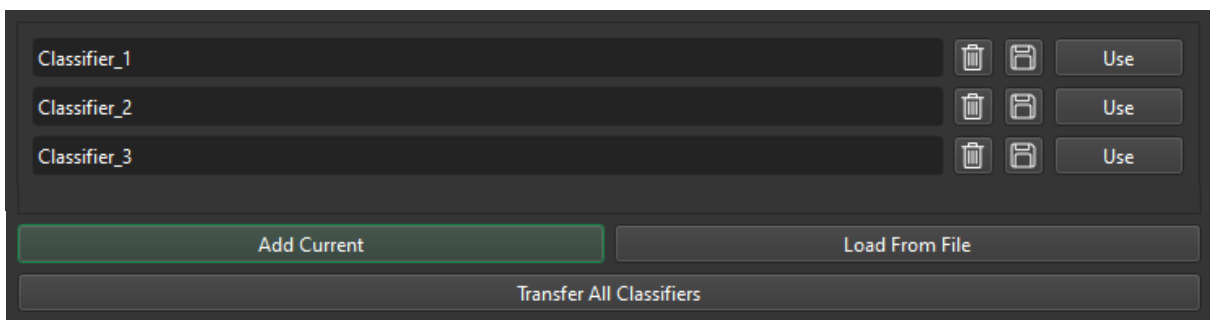


11.4.7 MULTIPLE CLASSIFIERS

The Multiple Classifiers feature provides the option to create multiple classifiers, transfer them to the camera, and run them in parallel.

1. Press the **Multiple Classifiers** button to open the corresponding window.
2. Clicking **Add Current** adds the currently loaded classifier to the list. Alternatively, an existing classifier can be added using **Load from File**.
 - a. It is recommended that you give it a unique name to make it easier to distinguish between classifiers.
 - b. Delete the classifier from the list by clicking the trash can icon or save the classifier to a separate file by clicking the save icon.
 - c. Press **Use** to activate the desired classifier in the 2D Cube View and the Canvas View. This allows you to switch back and forth between the classifiers in the list.

3. Closing the window allows you to train a new classifier or make changes. The list of already selected multiple classifiers remains available when you click Multiple Classifiers again.
 - a. **Please note that classifiers can only be added to a list if all settings for mean, background, and the classification algorithm are the same. Otherwise, an error message will be displayed.**
4. Once all relevant classifiers are added to the list, they can be transferred to the camera for live classification. In the **Waterfall** window under **Advanced** you can then switch between the individual classifiers. For more information about the on-camera classifier, see the chapter 11.5.



11.5 DATA ANALYSIS – ON CAMERA CLASSIFIER

The on-camera classifier is a functionality for the BlackIndustry SWIR cameras with firmware version 1.0.13 and later. This enables a previously trained classifier to be transferred to the camera so that newly captured data will be classified **on camera** in real time using a specific output mode. The classification results can be used directly in production without the need for additional hardware. In addition, the Ethernet bandwidth restriction and the performance overhead on the receiving device are eliminated, as the data from the camera only emerges classified.

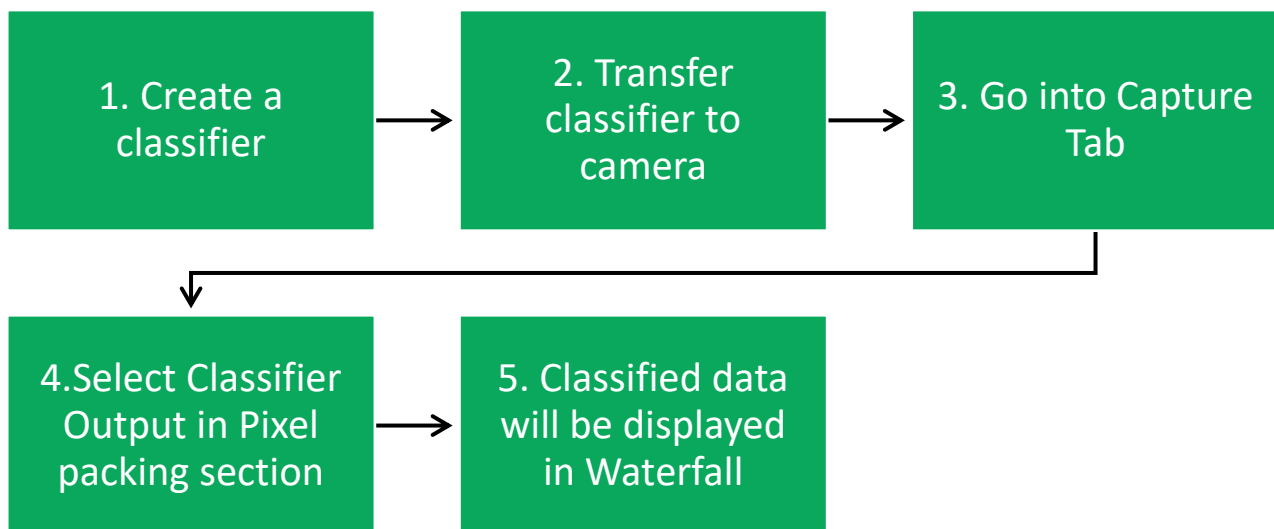
11.5.1 ON-CAMERA CLASSIFIER OUTPUT MODE

The following readout modes are available:

- **Classified Color:** The live classification results are displayed as an RGB visualization in the Waterfall Window, using the color scheme that was previously trained. No spectral information is transmitted, and the three other Live Windows are disabled.
- **Regression Float:** In addition to the classified color information, this mode returns the float values of the trained regression. These values cannot be displayed in BlackStudio, but can only be read out via API, for example.
- **Classified + Spectral:** The live classification results are displayed as an RGB visualization in the Waterfall Window, using the color scheme that was previously trained.

Simultaneously, the live spectral information is also transmitted, and the three other live windows are also active. However, when data is recorded using capture current waterfall or start/stop recording, only the spectral information is recorded, not the classified RGB information. Since spectral information is streamed together with classified data Ethernet bandwidth is a limiting factor. Therefore, this mode should only be used for debugging purposes.

11.5.2 ON-CAMERA CLASSIFIER IN FULL SPECTRAL RANGE:



9. Create a classifier according to the instructions in the chapter above.
10. Press **Transfer Classifier**. A new window opens:
 - b) Activate **Use Blackbalance** and/or **Use Whitebalance** to use the recorded Blackbalance or Whitebalance. If they are not used, either a suitable Blackbalance/Whitebalance already stored on the camera is used or an error screen appears after activating the Classified Color mode (see error screen below).
 - c) **Transfer Blackbalance/Whitebalance** to the connected camera. Previously saved balances will be overwritten.
 - d) Press **Take current camera values** and activate **Transfer FPS, Transfer Exposure** and **Transfer Gain**
 - e) Activate the checkbox **Transfer Classifier**.
 - f) Press **Transfer current to camera**. This transfers all activated settings to the camera, where they are stored until they are overwritten. Furthermore, the activated settings can be saved externally via **Export current into file** and reloaded from the file to the camera as often as desired.

☒ Use Blackbalance
☒ Use Whitebalance

☒ Transfer Blackbalance
☒ Transfer Whitebalance

 Black and Whitebalance will be used from the current selected mode:
 Mode 1: 952-1748nm, 1280x399, 2nm, 100FPS

 Transfer camera settings with classifier
☐ Transfer Regions
☐ Transfer FPS
☐ Transfer Exposure
☐ Transfer Gain

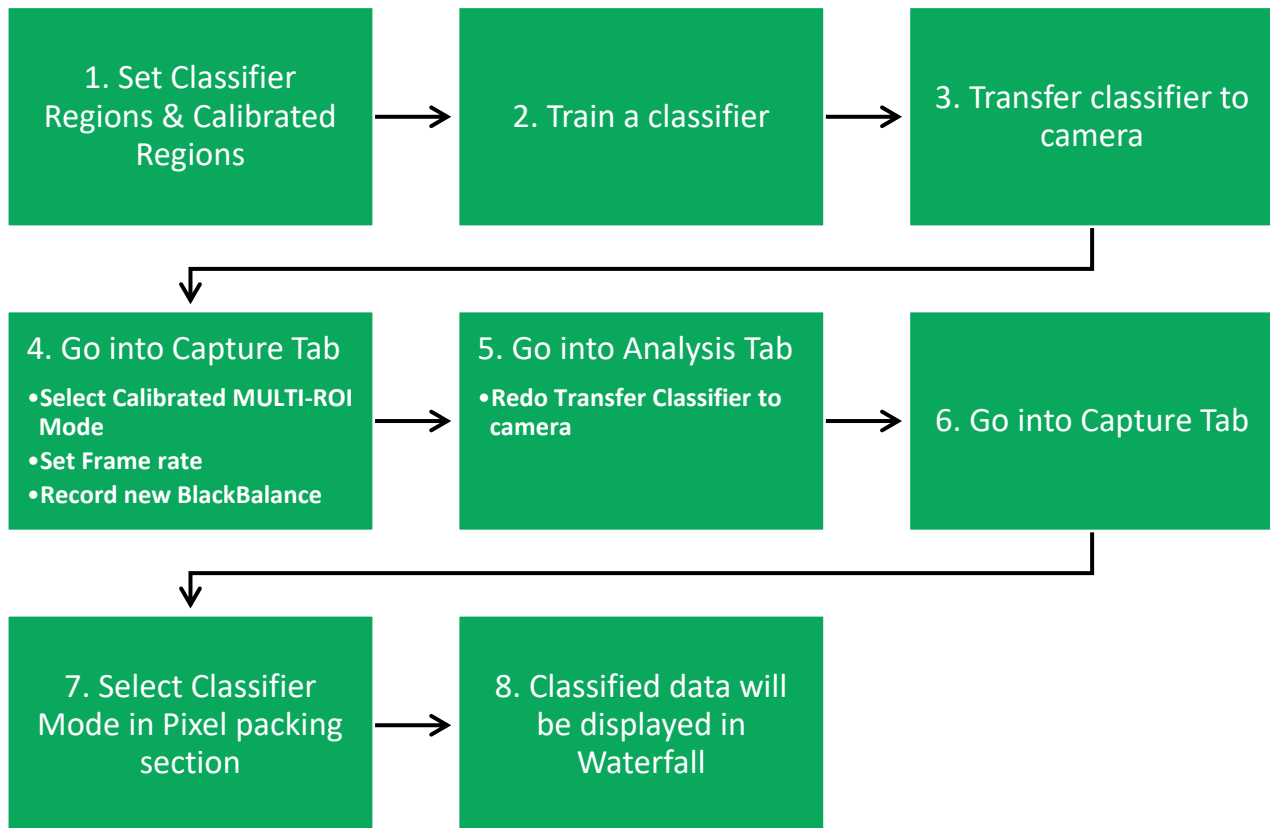
☒ Transfer Classifier

3. Go into the **Capture Tab** of BlackStudio.
4. Select **Classified Color** in the **Pixel Packing** section.

10 bit (2 byte)
 8 bit (1 byte)
 10 bit (5/4 byte)
 10 bit (3/2 byte)
 Classified Color
 Regression Floats

5. Newly acquired data will now be classified on camera and directly displayed in the **waterfall window** or can be received using the **API** or **GenICam**.

11.5.3 ON-CAMERA CLASSIFIER IN MULTI-ROI MODE USING SPECTRAL REGIONS:



1. If the **Calibrated Multi ROI mode** is to be used, the classifier can be configured using **Classifier Regions**.
 - a. These can be defined after enabling **Activate Classifier Regions**.
 - b. Press **Write Calibrated Regions to Camera**. This transfers the selected Spectral regions to the camera and saves them.
 - c. **Important:** The applied wavelength range (full or Spectral Regions) must always match the trained classifier. Otherwise, an error screen will be displayed when the **Classified Color mode** is activated (see below).
2. Create a classifier according to the instructions in the chapter above.
3. Press **Transfer Classifier**.
 - a. Activate **Use Blackbalance** and/or **Use Whitebalance** to use the recorded Blackbalance or Whitebalance. If they are not used, either a suitable Blackbalance/Whitebalance already stored on the camera is used or an error screen appears after activating the Classified Color mode (see error screen below).
 - b. **Transfer Blackbalance/Whitebalance** to the connected camera. Previously saved balances will be overwritten.
 - c. Press **Take current camera values** and activate **Transfer FPS, Transfer Exposure** and **Transfer Gain**
 - d. Activate **Transfer Regions**.
 - e. Activate the checkbox **Transfer Classifier**.

- f. Press **Transfer current to camera**. This transfers all activated settings to the camera, where they are stored until they are overwritten. Furthermore, the activated settings can be saved externally via **Export current into file** and reloaded from the file to the camera as often as desired.

☒ Use Blackbalance
☒ Use Whitebalance

☒ Transfer Blackbalance
☒ Transfer Whitebalance

 Black and Whitebalance will be used from the current selected mode:
 Mode 1: 952-1748nm, 1280x399, 2nm, 100FPS

 Transfer camera settings with classifier

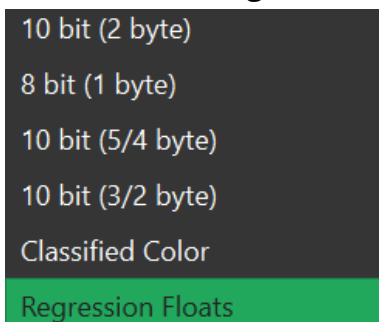
☒ Transfer Regions
☒ Transfer FPS
☒ Transfer Exposure
☒ Transfer Gain

☒ Transfer Classifier

Info: Classifier Regions will be used!
 The classifier will only work in Calibrated MULTI-ROI mode. Black and Whitebalance will be cut to these regions. Please set the Calibrated ROIs on camera to the identical values as for training the classifier. This can be done by setting them correctly. Then press the 'Set Classifier Regions on Camera' Button.

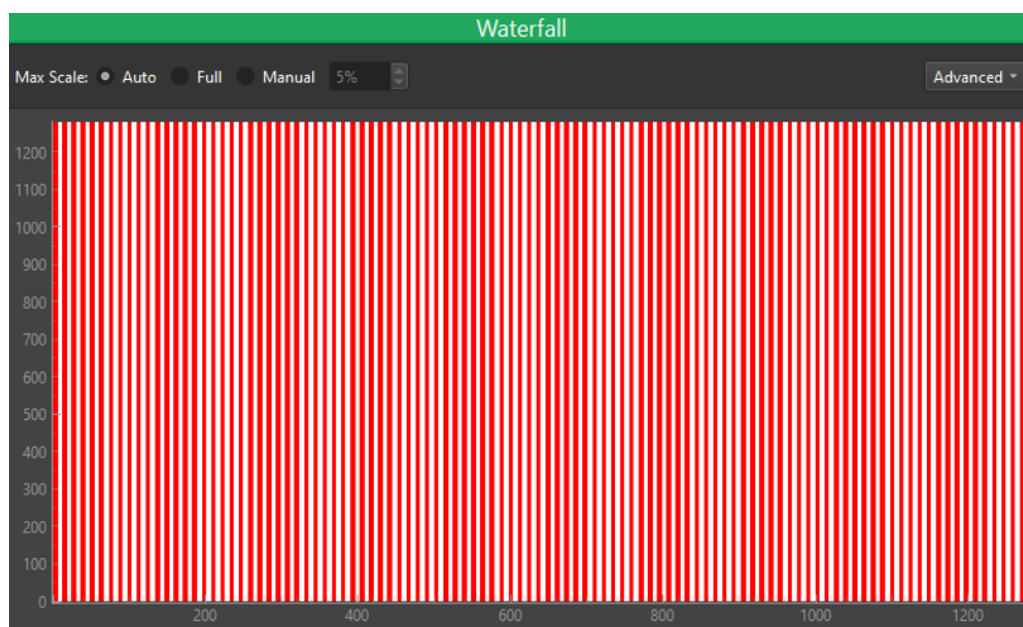
4. Go into the **Capture Tab** of BlackStudio.
 - a. Select the previously considered **camera mode** for which the classifier was trained (**Calibrated Multi-ROI mode**).
 - b. Set the desired **frame rate**.
 - c. Record a new **BlackBalance**. This must always be done when a new value is set for the frame rate.
5. Change into the **Analysis Tab** of BlackStudio.

- a. Press **Transfer Classifier** to transfer the Classifier including the newly acquired BlackBalance once again.
6. Change back into **Capture Tab** of BlackStudio.
7. Select a **Classifier Mode** in the **Pixel Packing** section:

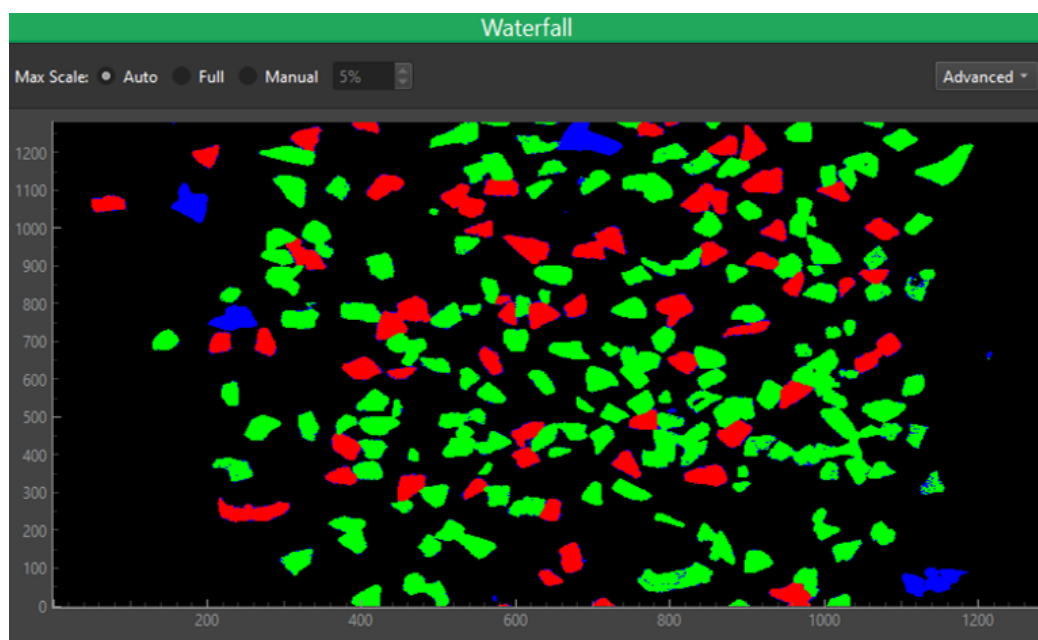


8. Newly acquired data will now be classified on camera and directly displayed in the **waterfall window** or can be received using the **API** or **GenICam**.
- ➔ Please consider that the use of the **Mean temporal filter** in the on-camera classifier increases the latency.
 - ➔ Please note that the spectral dimension of the classifier and the transferred White and Blackbalance must match the selected camera mode. If this is not the case, an **error screen** consisting of white and red stripes is transferred to the **Waterfall View** instead:

Error screen in Waterfall View:



Classified Waterfall window via on-camera classifier:



12 TROUBLESHOOTING

Rest assured, when it comes to errors or problems, we are committed to offering prompt assistance. Our dedicated team will work swiftly to address any issues you encounter.

We value your feedback! If you come across any issues or glitches with our product, please do not hesitate to get in touch. Your insights are invaluable in helping us improve and providing you with the best experience possible.

13 ACCESS TO HAIP BLACKSTUDIO SOFTWARE

You can download the newest HAIP BlackStudio software, including updates, and the corresponding user manual via the following link:

[BlackStudio Download](#) (Click here)

Or by scanning the QR-Code:



14 SAMPLE DATA

If you are interested in sample data for the products of HAIP Solutions GmbH, please follow the link below:

[Download Sample Data](#) (Click here)

Or scan the QR code:



15 MAINTENANCE AND SERVICE

HAIP Solutions GmbH is responsible for providing regular maintenance and updates to ensure the optimal functioning of the software. Maintenance activities may include bug fixes, security patches, performance optimizations, and compatibility enhancements. HAIP Solutions GmbH will strive to minimize any disruption to your system during maintenance activities and provide advance notice whenever possible.

We may offer technical support to assist you with any software-related inquiries or difficulties. It is recommended to upgrade to the latest supported version to continue receiving maintenance and support services.

16 SYSTEM REQUIREMENTS

Operating System	Windows 10 or later
Processor	Minimum Intel i5-1135G7 or equivalent
Memory (RAM)	Minimum 16 GB RAM
Storage Space	1 GB
Graphics Card	Integrated Graphic
Ports	1x GigE
Display	1920 x 1080 px or more

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